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Knowledge Gradient for Robust Selection of the Best

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We study sequential sampling for the robust selection-of-the-best (RSB) problem, where an uncertainty-averse decision maker facing input uncertainty aims to select from a finite set of alternatives via simulation the one with the best worst-case mean performance over an uncertainty set that consists of finitely many plausible input models. It is well known that the knowledge gradient (KG) policy is an efficient sampling scheme for the selection-of-the-best problem. However, we show that in the presence of input uncertainty a naïve but natural extension of the KG policy for the RSB problem is not convergent, i.e., fails to learn each alternative under each input model perfectly even with an infinite simulation budget. By reformulating the learning objective, we develop a so-called robust KG (RKG) policy for the RSB problem and establish its convergence, asymptotic optimality, and suboptimality bound. Due to its lack of analytical tractability, we approximate the RKG policy via Monte Carlo estimation and prove that the same asymptotic properties hold for the estimated policy as well. Numerical experiments show that the RKG policy outperforms several sampling policies significantly in terms of both normalized opportunity cost and probability of correct selection.

Key words: robust selection of the best; input uncertainty; knowledge gradient; sequential sampling

1. Introduction

Decision makers often encounter the problem of selecting the best from a finite set of alternatives, whose mean performances are unknown but can be estimated by running simulation experiments. For instance, a manufacturing manager may need to select a configuration of the production line to maximize the mean revenue, while an inventory manager may want to choose an inventory policy to minimize the total mean cost. This is known as the selection-of-the-best (SB) problem. To solve this problem, many selection procedures have been proposed either to determine the proper sample size of each alternative in order to provide certain statistical guarantee, or to allocate a limited number of opportunities for sampling across the alternatives in such a way as to maximize the information gained; see **?** and **?** for overviews.

In general, selection procedures for the SB problem are developed under the premise that the input model that drives the simulation experiment is given and fixed. Nevertheless, a decision maker often faces substantial *input uncertainty*, i.e., uncertainty about the input model, in many practical situations due to lack of input data; see Barton (2012) for a recent introduction.

Fan et al. (2013), **?**

In these examples, worst case analysis plays an important role because it helps people minimize the maximal expected loss. From the perspective of a rational decision maker, the strategy that minimize her cost over worst case is the best choice.

When the decision making function has no closed form, people can run simulations over the set of all feasible parameters for a great many of times and then select decision variable with the best average performance. However, this approach is not efficient since simulation comes with cost in reality and hence people need to balance the cost of simulations and precision on decision making.

It turns out that the construction of the optimal simulation policy is hard. In this paper, we study simulation polices which try to maximize decision making precision with fewest number of simulations and minimize their "distances" to the optimal one. We say a simulation policy is *robust* if it is "close" to the optimal policy.

1.1. Model Overview

We suppose that the unknown input distribution P is one of the elements in $\mathcal{P} = \{P_1, P_2, \dots P_K\}$ and set of all alternatives $\mathcal{S} = \{s_1, s_2, \dots s_M\}$ are given. Once a pair (P_i, s_j) is given, we can run simulation to sample the performance of alternative s_j with input distribution P_i. We call the pair (P_i, s_j) system (i, j) here. The cost of simulation may be expensive so simulation budget is tight. Then our goal is to design an efficient sampling policy that can help us to select the alternative in $\mathcal{S}$ that has best performance for the worst case over input distribution in $\mathcal{P}$ with high probability before the simulation budget is exhausted.

We assign multivariate normal prior to all the unknown expected performances of all systems in $\mathcal{P} \times \mathcal{S}$. We also assume that each simulation of system gives normal distributed unbiased random output with known variance. This setup is same as the one of the knowledge gradient (KG) policy in Frazier et al. (2009) and the merit of following it is that the posterior belief is also multivariate normal. Moreover, we assume independence among alternatives but allow correlations among expected performances of an alternative under different input distributions. That is, systems (i, j) and (k, l) are independent for any j, l if $i \neq k$. Under this assumption, simulation of an alternative under a certain input distribution gives information about its expected performances under different input distributions, but provides no information about other alternatives. Even though our theoretical analysis in later sections can be generalized to correlated cases, the correlation makes sparsity of

the prior covariance matrix vanish which leads to an significant increase of computational time complexity.

Suppose the simulation budget given is N then when we have run N simulations the final knowledge of all the systems can be denoted as (μ^N, Σ^N), which is the mean and covariance of a multivariate normal distribution. we call (μ^n, Σ^n) the nth stage of the sampling policy. We define an objective function with respect to the final stage (μ^N, Σ^N) and the optimization procedure of this objective function needs to be coincident with maximizing our knowledge about the correct alternative. Then we can turn our problem into a dynamic programming problem by designing a sampling policy that aims at maximize the objective function of (μ^N, Σ^N). This kind of problems are known as the *Markov Decision Problem*(MDP) from a general perspective. To construct the optimal policy, we need to solve the associated Bellman's equation backward in time, which is hard and has no closed form. Here, we consider some myopic policies for MDP and show that some of them are optimal in the infinite horizon case and perform very well in the finite horizon case.

1.2. Main Results

We list our main contributions as follows:

- We show that KG can be viewed as a generalized version of gradient descent algorithm from the perspective of dynamical system. Gradient descent is also a one-step optimal algorithm because. In general, choosing the simulation decision in each step can be viewed as choosing the steepest-descent direction. When some specific functions, for example non-concave function, are defined as the objective function, KG may "stick" at a local minimizer. As a result, KG prefers to stop running any simulation. This behavior of KG can be compared with gradient descent as follows. When we arrive at a local minimizer by running gradient descent iteration, we should stop any following iteration for iteration only leads to deviation from the minimizer. As a result, depending on the form of objective function, KG may not be asymptotically optimal which means as the number of simulations tends to infinity, the optimal objective value is achieved. In gradient descent, the same issue also occurs: as the number of gradient descent iteration tends to infinity, a local minimum other than a global one is returned.
- In order to study the simulation policy that minimize the uncertainty of the worst-case decision variable given finite simulation budget, we generalize the objective function of KG and run one-step optimal policy as our sampling policy. We call this policy *Robust Knowledge Gradient*(RKG). The decision making function of RKG has no closed form but with the helps from mathematical analysis tools, we can show that RKG preserve many KG's useful properties theoretically. Our analysis can be applied to more general object functions. More importantly, RKG indicates a way to modify objective function when simulation policy induced by the

original objective function fails to converge, in the sense of sampling each system infinitely often if an infinite simulation budget is available.

- In order to estimate the decision function of RKG, we introduce Monte Carlo(MC) estimation. We show the MC estimated RKG still preserves many important properties of RKG, for example convergence and asymptotic optimality. Further, we show a more interesting result: the gap between RKG and its MC estimator can be arbitrarily small for any simulation budget. More precisely speaking, since RKG is a one-step optimal policy, the gap between RKG and its MC estimator is greater than 0 in each round of simulation. We can show that as the number of simulations tends to infinity, the sum of these gaps is less than infinity in expectation. As a result, we can control the total estimation error by making the precision of every single estimation high enough. This is very crucial for two reasons. Firstly, the boundedness of error ensures that the "distance" between RKG and the MC estimator is bounded in infinite horizon case; secondly, MC estimated RKG is similar to stochastic gradient descent in the sense that even if random perturbation is introduced in each steepest descent, the gradient descent algorithm still converges to a local minimizer. These results further illustrate that any KG method is generalized gradient descent method.

Except for theoretical contributions, we also run several numerical experiments to show that our policy outperforms any other known policy obviously. In a standard Bayesian model, the performance of RKG has a higher order of Probability of Correct Selection(PCS) and a lower order of Normalized Opportunity Cost(NOC) compared with the second best policy. In real application, we run different policies to determine the worst-case decision variable of production line management and sS ordering policy. It turns out, under normal simulation budget, the PCS of RKG is almost 30% higher than the second best in production line management and 10% higher than the second best in sS. To summarize, RKG is the best known sampling policy on worst-case ranking and selection problems.

1.3. Literature Review

The R&S problem was first studied by Wald. Wald developed sequential analysis and Girshick apply it to the problem of ranking two alternatives. Based on the contribution of many forerunners, Bechhofer wrote down Bechhofer (1954) wrote down the formal definition of R&S problem. In R&S problem, n alternatives, each of which with distribution parameter θ_i, are given. Parameters of different alternatives are not necessarily the same. Then A random sample of size n is drawn from each alternatives. A statistical selection procedure uses this sample data to make a selection of alternatives in such a way that we can assert, with some specified level of confidence, that the

alternatives selected are the ones with the best reward. In the standard model of R&S problems, input distribution is known.

In the case of complex alternatives, simulation is needed for drawing random sample. Every simulation procedure measures the statistical error caused by sampling from the input models, typically via confidence intervals on the performance properties. However, these confidence intervals do not account for the possible misspecification of the input models when they are estimated from real-world data. Recently, many articles have shown that simulation error from input uncertainty can overwhelm the simulation sampling error(e.g. Barton,2012; Barton et al.,2014 and Chick 2001). This leads to unreliable results of stochastic R&S problems.

One approach to resolve the input uncertainty issue is Bayesian model averaging (BMA); please refer to Hoeting et al. (1999) for general tutorial and Chick (2001) for its application. Under the BMA framework, one assign prior probability to a set of candidate input distribution and take the average value as the input distribution.

Another approach in robust selection (Ben-Tal et al. 2009) is more appealing to decision maker when cost on implementing alternatives is high. This approach adopts worst-case scenario among the candidate input distributions to represent the value of a alternative. Motivated by it, Fan et al. (2013) modeled the input distribution uncertainty by a finite set of distribution and select the alternative having the best worst-case mean performance as the best one. They adopted a frequentest approach to analyze the pattern of data collected from samplings and make decision based on statistics and confidence interval.

A substantial amount of progress has been made using frequentist approaches in R$S problems. For example, Kim and Nelson (2001) and Kim and Nelson (2006) have presented policies which works quite well in the multistage setting with normal rewards. A general literature review of policies based on frequentist approaches may be found in Bechhofer et al. (1995).

Zhang and Ding (2016) followed the same modeling perspective, but they formulate problem in a Bayesian framework. One can take a Bayesian view on the true value of the system, which denotes the expected performance of an alternative under a candidate input distribution. Correlated prior belief to values are assigned to the true values and prior belief can be updated via simulations. When the simulation budget is exhausted, the final belief is used to select the alternative with best reword over worst case. The advantage of this approach is that Bayesian framework for R&S problem is well established to develop sequential sampling policies, including the optimal computing budget allocation (OCBA)(Chen et al. 1996, 2000, He et al. 2007), the knowledge gradient (KG) policy(Gupta and Miescke 1996, Frazier et al. 2008, 2009) and the expected value of information (EVI) approach (Chick and Inoue 2001a,b, Chick et al. 2010). The disadvantage is that, although OCBA, KG and VIP have been widely used for R&S, how to exploit R&S under input uncertainty

from a Bayesian framework is not well studied. In fact, the first Bayesian policy proposed by Zhang and Ding is not robust. They have modified their polices so that some of them have acceptable results in specific problem settings, but they did not provide any theoretical analysis in the article.

Indeed, most Bayesian experimental designs and Bayesian optimal learning problems have no optimal solution. A common suboptimal approach is to adopt a myopic one-step optimal policy (Gupta and Miescke, 1996; Chick et al., 2010; and Jones et al.,1998). Polices of this type are in the class of KG policies. The one-step optimal value function has closed form in most KG policies. To our knowledge, not much work has been done on studying how the policy is changed when MC method is applied to estimate the value function.

In this paper, we try to generalize KG and give a policy that can help people determine the decision variable under worst case.

The rest of the paper is organized as follows. We formulate the problem and introduce the Bayesian framework for robust R&S in §**??**. We begin the discussion of NKG in detail in §**??** by showing its non-convergence property. We will illustrate this property via a simple example. In §**??**, we introduce the new sampling scheme RKG and show its convergence, asymptotic optimality and suboptimality. The RKG policy is convergent; the RKG policy is optimal when there is only one simulation is given or there are infinitely many simulations are given; the sub-optimal gap of RKG is bounded in the finite sampling case. Then we will show that MC gives no effect to the convergence and asymptotic optimality of RKG and gives a small perturbation to its suboptimality. In §**??**, we present numerical experiments and demonstrate the excellent performance of RKG on two examples in production line setup and (s,S) order policy. In §**??**, we give conclusions. All the technical proofs are collected in the electronic companion.

2. Problem Formulation

In the setting of stochastic simulation, the performance measure of a simulation model is generally expressed as a function g of the decision variable s and the environmental variable ξ, where the former is controllable and deterministic whereas the latter is uncontrollable and random. The mean performance that we attempt to estimate via simulation is then

$$\mathbb{E}_P[g(s,\xi)],$$

where the expectation is taken with respect to ξ having probability distribution P. In the production line example s may be the capacity of each workstation, ξ may be the service rate, while g could be a revenue function of several variables of the system; in the (s,S) inventory example, s may be a vector (s_1, s_2) which indicates a (s, S) policy, ξ may be the demand rate of next period and g is the total cost per period.

Suppose that we have a set of M distinct possible decisions or alternatives $\mathcal{S} = \{s_1, \ldots, s_M\}$ and a set of K distinct possible distributions $\mathcal{P} = \{P_1, \ldots, P_K\}$. For a given distribution P, we define the optimal decision to be the one that delivers the smallest mean performance, i.e.,

$$\min_{s \in \mathcal{S}} \mathbb{E}_P[g(s, \xi)].$$

In light of the uncertainty about the distribution P, when assessing the decisions we adopt a robust perspective and base the comparison on the worst-case performance of a decision over the set $\mathcal{P}$. In particular, we are interested in the following optimization problem,

$$\min_{s \in \mathcal{S}} \max_{P \in \mathcal{P}} \mathbb{E}_P[g(s, \xi)]. \tag{1}$$

The most straightforward approach to estimate (1) is to run a large number of simulations on each pair $(s, P) \in \mathcal{S} \times \mathcal{P}$. This is inefficient for some alternates have mean performance far from the average and can be identified as being substantially better or substantially worse after a few simulations. Other alternatives may need more simulations before a precise decision can be made. Furthermore, the cost of each simulation could be expensive, and thus only few simulations are allowed. In the most extreme case, only one simulation is allowed so this approach is infeasible. From the Bayesian viewpoint, we can remove uncertainty about (1) after each simulation even if the simulation result is random. So we need to design a sequential sampling policy π aiming at minimizing the uncertainty of (1):

$$\min_{\pi} \ \textbf{uncertainty}(\min_{s \in \mathcal{S}} \max_{P \in \mathcal{P}} \mathbb{E}_P[g(s, \xi)]).$$

2.1. Bayesian Formulation

To facilitate the presentation, we refer to the pair (s_i, P_j) as "system (i, j)" and let $\theta_{i,j} = \mathbb{E}_{P_j}[g(s_i, \xi)]$, $i = 1, \ldots, M$, $j = 1, \ldots, K$. We let θ denote the matrix formed by the $\theta_{i,j}$'s and $\theta_{i:}^{\intercal}$ denote its ith row, i.e., $(\theta_{i,1}, \ldots, \theta_{i,K})$. Suppose that samples from system (i, j) are independent and have a normal distribution with *unknown* mean $\theta_{i,j}$ and *known* variance $\delta_{i,j}^2$. (In general, $g(s_i, \xi)$ is not normally distributed. Nevertheless, the sample average of a sufficiently large number of its multiple independent replications has approximately a normal distribution by the law of large numbers. We can view such a sample average as "one sample".)

Applying a Bayesian approach, we assume that the prior belief about θ is a multivariate normal distribution with mean μ^0 and covariance Σ^0, i.e., $\theta \sim \mathcal{N}(\mu^0, \Sigma^0)$, where Σ^0 is indexed by $((i, j), (i', j'))$, $1 \le i, i' \le M$, $1 \le j, j' \le K$. Further, we assume that the prior belief about θ is such that $\theta_{1:}, \ldots, \theta_{M:}$ are mutually independent and that the determinant $|\Sigma^0| > 0$. The reason we impose such a constraint on Σ^0 is because we need to rule out cases that a subset of systems is perfectly

correlated to another disjoint subset. When two sets are perfectly correlated to each other, knowing the vlaues on one set gives you full information about the values of the other. If one encounters the case $|\Sigma^0| = 0$ then she can simply take off one of the perfectly correlated set and spend sampling budget on the rest so that the new Σ^0 has determinant greater than 0.

Consider a sequence of N sampling decisions, $(x^0, y^0), (x^1, y^1), \ldots, (x^{N-1}, y^{N-1})$. At each time $0 \le n < N$, the sampling decision (x^n, y^n) selects a system from the set $\{(i,j) : 1 \le i \le M, 1 \le j \le K\}$. Conditionally on the decision (x^n, y^n), the sample observation is $\hat{z}^{n+1} = \theta_{x^n,y^n} + \epsilon^{n+1}$, where $\epsilon^{n+1} \sim \mathcal{N}(0, \delta^2_{x^n,y^n})$ is the sampling error. We assume that the errors $\epsilon^1, \ldots, \epsilon^N$ are mutually independent and are independent of θ.

We define a filtration $\{\mathcal{F}^n : 0 \le n < N\}$, where $\mathcal{F}^n$ is the sigma-algebra generated by the samples observed and the decisions made by time n, namely, $(x^0, y^0),\ \hat{z}^1, \ldots,$
$(x^{n-1}, y^{n-1}),\ \hat{z}^n$. We use $\mathbb{E}_n[\cdot]$ to denote the conditional expectation $E[\cdot|\mathcal{F}^n]$ and define $\mu^n := \mathbb{E}_n[\theta]$ and $\Sigma^n := Cov[\theta|\mathcal{F}^n]$. By Bayes rule, the posterior distribution of θ conditionally on $\mathcal{F}^n$ is multivariate normal with mean μ^n and covariance Σ^n. Our uncertainty about θ decreases during the process of the sequential sampling. After all the N sampling decisions are executed, the decision-maker selects a system that attains $\min_i \max_j \mu^N_{i,j}$ in light of (1).

Intuitively, the process of sequential sampling can be viewed as a learning process that removes the randomness of true underlying value θ. In fact, μ^n converges to θ almost surely fast if a convergent policy is applied, which is an obvious result derived from the *strong law of large numbers*.

We now can show how μ^{n+1} and Σ^{n+1} in terms of μ^n, Σ^n, (x^n, y^n) and $\hat{z}^{n+1}$. The independence assumption on $\theta_{x:}$ and $\theta_{x':}$ gives:

$$\Sigma^n_{x:,x':} = \mathbf{0}, \quad \text{if } x \ne x',$$

for all $0 \le n < N$, where $\Sigma^n_{x:,x':}$ denotes the covariance matrix of $\theta_{x:}$ and $\theta_{x':}$ conditionally on $\mathcal{F}^n$. Sampling system (x, y) provides no information about system (x', y') if $x' \ne x$.

Then we can use Bayes' rule (see Gelman et al.. 2004) and apply the Sherman- Woodbury matrix identity (see Golub and Van Loan. 1996) to obtain the recursions:

$$\mu^{n+1}_{x:} = \begin{cases} \mu^n_{x:} + \dfrac{\hat{z}^{n+1} - \mu^n_{x,y}}{\delta^2_{x,y} + \Sigma^n_{(x,y),(x,y)}} \Sigma^n_{x:,x:} e_y, & \text{if } x^n = x,\ y^n = y, \\ \mu^n_{x:}, & \text{if } x^n \ne x,\ y^n = y, \end{cases} \tag{2}$$

and

$$\Sigma^{n+1}_{x:,x:} = \begin{cases} \Sigma^n_{x:,x:} - \dfrac{\Sigma^n_{x:,x:} e_y e_y^\intercal \Sigma^n_{x:,x:}}{\delta^2_{x,y} + \Sigma^n_{(x,y),(x,y)}}, & \text{if } x^n = x,\ y^n = y, \\ \Sigma^n_{x:,x:}, & \text{if } x^n \ne x,\ y^n = y, \end{cases}$$

where e_y is a vector in $\mathbb{R}^K$ whose elements are all 0's except a single 1 at index y.

We now define a $\mathbb{R}^K$-valued function $\tilde{\sigma}$ as

$$\tilde{\sigma}(\Sigma, x, y) := \frac{\Sigma_{x:,x:} e_y}{\sqrt{\delta^2_{x,y} + \Sigma_{(x,y),(x,y)}}}, \tag{3}$$

and we define a random variable Z^{n+1} as

$$Z^{n+1} := \frac{\hat{z}^{n+1} - \mu^n_{x^n,y^n}}{\sqrt{\delta^2_{x^n,y^n} + \Sigma^n_{(x^n,y^n),(x^n,y^n)}}}.$$

Then Z^{n+1} is standard normal conditionally on $\mathcal{F}^n$, since

$$var[\hat{z}^{n+1} - \mu^n_{x^n,y^n} | \mathcal{F}^n] = var[\theta_{x^n,y^n} + \epsilon^{n+1} | \mathcal{F}^n] = \delta^2_{x^n,y^n} + \Sigma^n_{x^n:,x^n:}.$$

It follows from (2) and (3) that

$$\mu^{n+1}_{x:} = \begin{cases} \mu^n_{x:} + \tilde{\sigma}(\Sigma^n, x^n, y^n) Z^{n+1}, & \text{if } x^n = x,\ y^n = y, \\ \mu^n_{x:}, & \text{if } x^n \neq x,\ y^n = y. \end{cases} \tag{4}$$

We note from (2.1) that the determinant of Σ^n is decreasing. This can be interpreted by thinking of the uncertainty of θ is decreasing. The sampling result at time n removes some of its uncertainty.

2.2. Dynamic Programming

We assume that each sampling decision (s^n_i, P^n_j) is taken over the finite set $\mathcal{S} \times \mathcal{P}$ and these decision are made sequentially, in that (s^n_i, P^n_j) is allowed to depend on samples observed by time n. So we can have $(s^n_i, P^n_j) \in \mathcal{F}^n$. We define Π to be the set of feasible sampling order satisfying the sequential requirement. That is, Π is the space of feasible adapted policies defined as

$$\Pi := \{\big((s^0_i, P^0_j), \ldots, (s^{N-1}_i, P^{N-1}_j)\big) : \ (s^n_i, P^n_j) \in \mathcal{F}^n\}.$$

We will use π to denote a generic element in Π and use $\mathbb{E}^\pi[\cdot]$ to indicate expectation taken when the measurement policy is fixed to π. Our goal is to choose a sampling policy minimizing expected cost over worst case. The objective function of a naive extension of KG can be written as:

$$\min_{\pi \in \Pi} \mathbb{E}^\pi \left[\min_{1 \le i \le M} \max_{1 \le j \le K} \mu^N_{i,j} \right]. \tag{5}$$

Here we view $\max_{1 \le j \le K} \mu^N_{i,j}$ as the worst-case performance of alternative i. However, this objective function leads to a non-convergent KG policy. It is mainly for two reasons: first, the expected worst case performance is not equivalent to the worst among expected performances; second, bounded supper-martingale structure needed by KG vanishes under such framework. As we will see later, the bounded supper-martingale structure is the essential of KG policy, which guarantees its convergence property.

So we rewrite the objective function as:

$$\min_{\pi\in\Pi}\mathbb{E}^{\pi}\left[\min_{1\le i\le M}\mathbb{E}[\max_{1\le j\le K}\theta_{i,j}|\mathcal{F}^N]\right]. \tag{6}$$

Compared with (5), this objective preserves the supper-martingale structure which we will give fully discussion in the appendix.

In general, given a function f of state $s=(\mu,\Sigma)$, we can define an objective function

$$\min_{\pi\in\Pi}\mathbb{E}^{\pi}\left[f(\mu^N,\Sigma^N)\right]. \tag{7}$$

Clearly, μ^n takes its values in $\mathbb{R}^{M\times K}$ while Σ^n is in the space of positive semidefinite matrices of size $(MK)\times(MK)$. We define $\mathbb{S}$, the state space of $S^n:=(\mu^n,\Sigma^n)$, to be the cross-product of these two spaces. In our framework, $\forall s=(\mu,\Sigma)\in\mathbb{S}$, $||\mu||<\infty$ and $0<|\Sigma|<\infty$. So $\mathbb{S}$ is open. Define the value function $V^n:\mathbb{S}\mapsto\mathbb{R}$

$$V^n(s):=\min_{\pi\in\Pi}\mathbb{E}^{\pi}\left[f(S^N)\middle|S^n=s\right],\quad s\in\mathbb{S}.$$

Then, the terminal value function is given by

$$V^N(s)=f(s),\quad s=(\mu,\Sigma)\in\mathbb{S},$$

and our goal is to compute $V^0(s)$ for any $s\in\mathbb{S}$. The dynamic programming principle dictates that the optimal value function $V^n(s)$, for any $0\le n<N$, can be computed by recursively solving

$$V^n(s)=\min_{1\le x\le M,1\le y\le K}\mathbb{E}\left[V^{n+1}(S^{n+1})\middle|S^n=s,(x^n,y^n)=(x,y)\right]. \tag{8}$$

Then the Q-factors, $Q^n:\mathbb{S}\times\{1,...,M\}\times\{1,....K\}\to\mathbb{R}$ is defined as:

$$Q^n(s,(x,y)):=\mathbb{E}\left[V^{n+1}(S^{n+1})\middle|S^n=s,(x^n,y^n)=(x,y)\right]$$

The Q-factor $Q^n(s,(x,y))$ can be thought of as giving the value of being in state s at time n, sampling from alternative (x,y), and then behaving optimally afterward. We let $\mathcal{A}^{n,\pi}:\mathbb{S}\mapsto\{1,\ldots,M\}\times\{1,\ldots,K\}$ be the function that satisfies $\mathcal{A}^{n,\pi}(S^n)=(x^n,y^n)$ almost surely under the probability measure $\mathbb{P}^{\pi}$, which is induced by a Markovian policy π, and call this function the decision function for π. A policy is said to be stationary if $\mathcal{A}^{n,\pi}$ in dependent of n, i.e. $\mathcal{A}^{0,\pi}=\mathcal{A}^{1,\pi}=\ldots\mathcal{A}^{N-1,\pi}$ almost surely under $\mathbb{P}^{\pi}$. Furthermore, we simply write $\mathcal{A}^{\pi}$ if π is stationary. We define the value function for a policy π as:

$$V^{n,\pi}(s):=\mathbb{E}^{\pi}\left[f(S^N)\middle|S^n=s\right]$$

The dynamic programming principle states that any policy π with sampling selection:

$$\mathcal{A}^{n,\pi}(s)\in\operatorname*{arg\,min}_{1\le x\le M,1\le y\le K}Q^n(s,(x,y))$$

is optimal.

Given the sample size N, we use $V^n(\cdot, N) : \mathbb{S} \mapsto \mathbb{R}$ to denote the optimal value function at time n. Similarly, $V^{n,\pi}(\cdot, N) : \mathbb{S} \mapsto \mathbb{R}$ denotes the value function of policy π at time n when the terminal time is N.

3. Naive Knowledge Gradient

We define the NKG policy π^{NKG} to be the stationary policy with decision function:

$$\mathcal{A}^{\pi^{\text{NKG}}}(s) = \underset{1\le x\le M, 1\le y\le K}{\arg\min} \left\{ \mathbb{E}_n \left[\min_{1\le i\le M} \max_{1\le j\le K} \mu_{i,j}^{n+1} \middle| S^n = s, (x^n, y^n) = (x, y) \right] - \min_{1\le i\le M} \max_{1\le j\le K} \mu_{i,j}^n \right\}.$$

To compute the above decision function, the key step is to compute the expectation inside the curly braces. Note that by (4), $\mu_{i,j}^{n+1}$ is a linear transform of the same standard normal random variable Z^{n+1} for all (i, j)'s. This expectation can be expressed in the form of $\sum_k \mathbb{E}([a + bZ)\mathbb{I}_{c_k \le Z < c_{k+1}}]$, for some constants a, b, and c_k's. The sequence of c_k's is in fact the change points of a piecewise linear function, formed by the minimum of the M maxima of linear functions that transform $\mu_{i,j}^n$ to $\mu_{i,j}^{n+1}$. These change points can be computed by a sweep line algorithm combined with a divide-and-conquer strategy; see Section 6.2.1 of Sharir and Agarwal (1995) for details of such an algorithm.

Note that if $K = 1$, then NKG is reduced to KG. The name KG stems from the following observation: $\min_i \max_j \mu_{i,j}^{n+1} - \min_i \max_j \mu_{x,j}^n$ may be thought of as a gradient in some sense since it represents the incremental random value of the sampling decision (x, y) at time n.

3.1. Non-Convergence Result

Before the study of convergence property, we need to give a strict definition of convergence. Given a covariance matrix $\Sigma \in \mathbb{R}^{d\times d}$, we first define its operator norm:

$$||\Sigma|| := \sup_{V\in\mathbb{R}^d : ||V||_2 = 1} ||\Sigma V||_2.$$

We can derive that if $||\Sigma|| = 0$ then $|\Sigma| = 0$ but the reverse direction is not true. Given a set of systems, if our prior $||\Sigma^0|| = 0$ then we have perfect information so we do not need to do any sampling. On the other hand, if $|\Sigma^0| = 0$, $||\Sigma^0||$ is not necessarily equal 0. $|\Sigma| = 0$ only tells that some subsets of systems are perfectly correlated. In our framework, we have assume that $|\Sigma^0| > 0$ so $||\Sigma^0|| > 0$. We first state the following theorem:

THEOREM 1. *Given a ploicy π, if $|\Sigma^0| > 0$, sampling budget is infinite and each sampling comes with noise then the following four statements are equivalent:*

1. *every system is sampled infinitely often under π.*

2. $\lim_{n\to\infty} ||\Sigma^n|| = 0$ *under* π.
3. *the probability that policy* π *identifies the system which satisfies any objective requirement tends to 1 as number of samplings tends to infinity.*
4. $\lim_{n\to\infty}(\mu^n, \Sigma^n) = (\theta, 0)$ *under* π

The proof is left in appendix. We say a policy π is convergent if it satisfies one of the four conditions mentioned above and hence the meaning of convergence is interchangable. Note that if we do not assume that $|\Sigma^0| > 0$, then condition 1 is not equivalent to 2, 3 and 4 but 2, 3 and 4 are still equivalent in this case.

It is shown in Frazier et al. (2008) that KG is *convergent*, in the sense that it eventually identifies the system that is truly the optimal given sufficient computational budget. However, NKG, as a naive extension of KG to the setting of $K \geq 2$, is not convergent in general.

Convergence of a policy on its own indicates little about efficiency of the policy in the finite sample case. For instance, the equal allocation policy which allocates the computational budget in a round-robin fashion equally among the systems guarantees that every system is sampled infinitely often if infinite computational budget is available, and thus it is convergent. But its performance in the finite sample case is not particularly satisfying. Nevertheless, convergence should be a desired feature of a good sampling policy as it ensures that the policy does not "stick" in a proper subset of the systems, in which case the other systems would not be sampled infinitely often and thus would never be learned perfectly even given infinite computational budget.

We discuss intuition of the non-convergence property and use steepest gradient descent as an analogy here, leaving proofs in our appendix.

Given a differentiable function $f : \mathbb{R}^m \to \mathbb{R}$, if our goal is to search $s^* = \arg\min_s f(s)$, we can set $s^n = s^{n-1} - \epsilon \bigtriangledown f(s^{n-1})$ for any $s^0 \in \mathbb{R}^n$ and $\epsilon > 0$ and run the iteration until s^n converges. Then we get a local minimizer $s^* = \lim_{n\to\infty} s^n$, where $f(s^*) \leq f(s)$ for any s in a neighborhood of s^*. Now suppose that every update is restricted on only m directions. For example, let e_x be a vector in $\mathbb{R}^m$ of 0's with a single 1 at index $x \in \{1, 2, ..., m\}$ and our goal is to update one of $\{s^n_x\}_{x=1}^m$ in each iteration n so that s^n converges to a local minimizer. In this case, we can simply select $x^* = \arg\max_x \frac{\partial f(s)}{\partial s_x}\big|_{s^{n-1}}$ and let $s^n = s^{n-1} - \epsilon \frac{\partial f(s)}{\partial s_{x^*}}\big|_{s^{n-1}} e_{x^*}$. When $n \to \infty$, s^n converges to a local minimizer s^* of f.

From the perspective of dynamical system, we can view $\{s^n\}_{n=0}^\infty$ as a dynamics at initial state s^0 and s^* as an attractor, meaning that starting at any initial state s^0 near s^*, the dynamics $\{s^n\}$ converges to s^* and, more importantly, if $s^0 = s^*$, then $s^n = s_0$ for any n.

In the frame work of knowledge gradient, we can also view the evolution of S^n as a dynamics. Given a policy π, the evolution of $\{S^n\}$ is governed by $S^{n+1} = T(S^n, Z, (x^\pi(S^n), y^\pi(S^n))) = g(S^n, Z)$, where $Z \sim \mathcal{N}(0, 1)$ and T and g are some state transition functions.

Now given a function f, a set of sampling decisions with indexes $\{1,2,...,m\}$ and initial state $S^0 = s$, we run a one-step optimal policy π:

$$x^\pi(S^n) = \underset{x\in\{1,2,...,m\}}{\arg\min}\ \mathbb{E}[f(S^{n+1})|S^n,\ x^n = x].$$

Suppose all the m systems on which we take sample are independent, then $S^{n+1} = \big(\mu^n + (\mu^{n+1} - \mu^n)e_x,\ \sigma^n + (\sigma^{n+1} - \sigma^n)e_x\big)$. This is similar to the steepest gradient descent restricted on m directions. That is, the choice which gives the greatest improvement is selected. We say that $\{S^n\}$ is a random dynamics induced by π and for any $s \in \mathbb{S}$, we say that s is an attractor if $S^n = s$ and the choice of x^n induced by π is such that S^{n+1} is close to s as much as possible in expectation. For formal definition of random attractor, please refer to Arnold (1998).

An attractor s tells that the policy π tends to stay near s so π denies all the sampling decision x^* which very likely leads to a new state far from s. Moreover, if the variance is large, then any sampling decision will give a new updated state far from s so it is very likely that an attractor $s = (\mu, \Sigma)$ is with small $|\Sigma|$. We called the connected set of attractors a *trap*, which is equivalent to absorbing set in the study of random dynamical systems. If the measure of a trap T, which is defined on $\mathbb{S}$, is greater than 0, then the probability that π fails to converge is also greater than 0 for the following reasons. If $S^n \in T$, then π will omits the sampling choices with low probability that $S^{n+1} \in T$ and in the next round, if $S^{n+1} \in T$, the choices omitted in the previous round will be omitted again. On the other hand, the system, say system x, sampled in the previous round will be selected again because it is with smaller variance compared with previous round. As we take increasingly more samples on x, $S^n \in T$ with increasingly higher probability and $\mathbb{P}(S^\infty \in T) > 0$ as a result.

A special case needs to be noticed is that:

$$f(s) \le \mathbb{E}[f(S^{n+1})|S^n = s, x^n = x], \qquad \forall\ x \in \{1,2,...,m\}$$

with $s = (\mu, \Sigma)$, $|\Sigma| > 0$. This means sampling only leads to worse result and the best thing to do is to stop sampling. We can compare this case to gradient descent: when s^n is at a local minimizer, the best thing to do is to stop the iteration. In fact, this is what happens to NKG as we will show in the appendix.

We also notice that for any convergent policy π, only one attractor exists for the random dynamics $\{S^n\}$ induced by π, namely $(\mu^\infty, 0)$.

Here, we give two examples to illustrate attractor and trap. The first one is a simple example which states that if we apply one-step optimal algorithm to minimize $||\mu^N||^2$, then any $(\mu^\infty, \sigma^0 - e_x\sigma^0_x)$ is an attractor for any x and any σ^0:

EXAMPLE 1. Let $f(s)=\mu^T\mu$, μ_i^n is independent of μ_j^n if $i\neq j$ and $\pi(s)=\arg\min_x \mathbb{E}[f(\mu^{n+1})|S^n=s,x^n=x]$. Then $\pi(s)=\arg\min_x Var(\mu^{n+1}|S^n=s,x^n=x)$ for any $s\in\mathbb{S}$. Previous sampling decision $x^{n-1}=x$ leads to smaller $Var(\mu^{n+1}|S^n,x^n=x)$. Therefore, π gives identical sampling decision from the very beginning. Therefore, $(\mu^\infty,\sigma^0-e_x\sigma_x^0)$ is an attractor for any x and any σ^0.

The second example shows that if we apply one-step optimal policy on $\min\mathbb{E}[f(\mu^N)]$, where f is a differentiable function with positive value, then the local minimizer of f is contained in a trap.

EXAMPLE 2. We assume θ_i is independent of θ_j if $i\neq j$; $f(s)=f(\mu)>0$ for any $s=(\mu,\sigma_1,\sigma_2,...)\in\mathbb{S}$; μ^* is a local minimizer of f. Then we have:

$$\mathbb{E}[f(\mu^{n+1})|\mu^n=\mu^*,x^n=x]=\frac{1}{\sqrt{2\pi}\sigma_x}\int f(\mu_1^*,...,\mu_{x-1}^*,t,...)\exp\{-\frac{1}{2}\frac{(t-\mu_x^*)^2}{\sigma_x^2}\}dt.$$

This equation is smooth in σ_x and converges to $f(\mu^*)$ from above as $\epsilon>\sigma_x\to 0$ for some small $\epsilon>0$. If $\max_x\sigma_x$ is small enough, $\pi^{KG}(\mu^*,\sigma)=\arg\min_x\sigma_x$ for choosing the smallest σ_x means putting more weight on $f(\mu^*)$. As a result, (μ^*,σ) is an attractor for any σ small enough. We then define function

$$h_x(\mu,\sigma)=\frac{1}{\sqrt{2\pi}\sigma}\int f(\mu_1,...,\mu_{x-1},t,...)\exp\{-\frac{1}{2}\frac{(t-\mu)^2}{\sigma}\}dt.$$

From basic calculation, we can derive that $\partial_\sigma h_x(\mu^*,\sigma)\neq 0$ for any small σ and any x. Then according to implicit function theorem, we have a neighborhood N of μ^* and a function g such that $h_x(\mu,g(\mu))=h_x(\mu^*,\sigma)$ for any $\mu\in N$ and any x. So $\{(N,\sigma)|\sigma<\epsilon\}$, for some $\epsilon>0$, is a trap according to definition.

On the contrary, if $f=f(\mu)$ is a concave function, one-step optimal policy is always convergent. A concave(convex) function can "capture" uncertainty. A simple case is the Jensen's inequality: $\mathbb{E}[f(X)]\leq f(\mathbb{E}X)$ if f is concave. To be more precise, given a concave function f and state $s=(\mu,\Sigma)$, we can have:

$$\mathbb{E}^n\left[f(\mu^{n+1})\middle|S^n=s,x^n=x\right]-f(\mu^n)=tr[\nabla^2 f(\mu)Cov(\mu^{n+1})]+o(|Cov(\mu^{n+1})|).$$

The second term on the right hand side is a perturbation of smaller order; the first term, which is always non-positive, indicates the uncertainty removed if sampling decision x is chosen. When $f(x_1,\ldots,x_n)=\min\{x_1,\ldots,x_n\}$, we also have similar result by using the concepts of sub-hessian and sub-gradient (see Scheimberg and Oliveira, 1992); In fact, $\min\{\mu^n\}-\mathbb{E}^n[\min\{\mu^{n+1}\}]$ is positively correlated to $|Cov(\mu^{n+1})|$ when μ^{n+1} is normal distributed (Ross 2010). As a result, the improvement $f(\mu^n)-\mathbb{E}^n[f(\mu^{n+1})]$ in each step is always positive and can be interpreted as the amount of "uncertainty" removed. In this case, S^{n+1} is trying to stay far from S^n in each iteration so the only attractor is $(\mu^\infty,0)$ in these cases.

However, in the appendix, we will show that if $f(.) = \min\{\max\{\cdot\}\}$, given any μ and small Σ, (μ, Σ) is an attractor. As the number of sampling increases, S^n must be in an attractor for any n large enough so NKG will omit all the decision with relatively large uncertainty, resulting in non-convergent sampling order. By concluding what are mentioned above, we have the theorem:

THEOREM 2. *Under NKG policy, $\exists N^* < \infty$ a.s. and i^* such that $x^n \neq i^*$ for all $n \geq N^*$.*

The general idea of the proof for theorem (2) is as follows. The order of $\{\max_j \mu_{ij}^n\}_{i=1}^M$ stops changing after finite number of sampling. When the order of $\{\max_j \mu_{ij}^n\}_{i=1}^M$ stops changing, sampling some systems provides no improvement due to the non-convexity of $\min\{\max\{\cdot\}\}$. As a result, NKG will never sample those systems and any sate s becomes an attractor in this stage.

3.2. A Concrete Example

We can further demonstrate the non-convergence of NKG via the following special case. More realistic numerical experiment will be shown in later section.

EXAMPLE 3. Let $K = 2$. Suppose that every element of Σ^0 equal to 0 except $\Sigma^0_{(i,j),(i,j)}$ is clsoe to 0 for any $(i,j) \neq (1,1)$ and $\Sigma^0_{(1,1),(1,1)} >> 0$. In other words, the prior belief about θ is such that $\theta_{1,1}$ is with relatively high randomness, whereas $\theta_{1,2}, \theta_{2,1}, \theta_{2,2}$ are with relatively small randomness.

We can further assume that every element of Σ^0 equal to 0 except $\Sigma^0_{(1,1),(1,1)} > 0$ even if this assumption violates our framework. We can build the legal example on this assumption later with ease.

The updating equation (4) implies that if $(x^0, y^0) = (1,1)$, then

$$\mu^1_{1,1} = \mu^0_{1,1} + \sigma Z, \quad \text{and} \quad \mu^1_{i,j} = \mu^0_{i,j},\ (i,j) \neq (1,1),$$

for some $\sigma > 0$, where Z is a standard normal random variable; otherwise, $\mu^1_{i,j} = \mu^0_{i,j}$ for any (i,j).

Clearly, the expected single-period reward associated with the sampling decision (i,j) is 0 if $(i,j) \neq (1,1)$. With $(x^0, y^0) = (1,1)$, the same quantity becomes

$$\mathbb{E}\left[\max\left(\mu^0_{1,1} + \sigma Z^1, \mu^0_{1,2}\right) \wedge \max(\mu^0_{2,1}, \mu^0_{2,2})\right] - \min_i \max_j \mu^0_{i,j}. \tag{9}$$

Without loss of generality, set $\mu^0_{1,1} = 0$. Consider the special case where

$$\mu^0_{1,2} < 0 < \max(\mu^0_{2,1}, \mu^0_{2,2}).$$

It follows that $\min_i \max_j \mu^0_{i,j} = 0$ and that (9) equals

$$a\mathbb{P}(\sigma Z < a) + b\mathbb{P}(\sigma Z > b) + \mathbb{E}[\sigma Z \mathbb{I}_{a \leq \sigma Z \leq b}], \tag{10}$$

where $a = \mu^0_{1,2}$ and $b = \max(\mu^0_{2,1}, \mu^0_{2,2})$, since a and b are both constants. It is easy to show that (10) is negative if $a + b < 0$. Hence, the optimal decision is not to sample the unknown $\theta_{1,1}$ but to sample any of the known systems, in which case the state of the systems remains the same in all the subsequent time epochs. Consequently, if NKG is adopted, system (1,1) will never be sampled.

Now, we can put small value on $\Sigma^0_{(i,j),(i,j)}$ for each $(i,j) \neq (1,1)$. If all the values are small enough, then all previous equations are perturbed slightly. Therefore, system (1,1) will never be sampled still.

By contrast, if $M = 1$, then it is equivalent to set $b = \infty$ in (10) and the expected single-period reward is always positive if the decision is to sample system $(1,1)$. So the same policy would encourage exploration of uncertainty rather than discourage it, thereby being convergent.

In fact, if we drop the condition that $|\Sigma^0| > 0$, we can directly assume that $\Sigma^0_{(i,j),(i,j)} = 0$ except $\Sigma^0_{(1,1),(1,1)}$. Then NKG still violates condition 2,3 and 4 in theorem 1 . From this perspective, NKG is still non-convergent.

The non-convergence of NKG is not surprising if KG is related to ordinary gradient descent method. The original objective function: $\max_\pi E[\max_i \mu^N_i]$ of KG is of some monotone and convex properties shown by Frazier et al. (2008) leading to the convergence result. However, if we put a min in between making the objective function becomes $\max_\pi E[\min_i \max_j \mu^N_{ij}]$ or $\max_\pi \min_i E[\max_j \mu^N_{ij}]$, those properties are lost. In ordinary gradient descent method, if the objective function is convex or monotone, global extrema is guaranteed when number of iteration is infinite. When the object function is non-convex, only local extrema is guaranteed by gradient descent method. So we modify our objective functions that necessary properties for convergence remain.

4. Robust Knowledge Gradient

We define the RKG policy π^{RKG} to be the stationary policy with decision function:

$$\begin{aligned}
&\mathcal{A}^{\pi^{\mathrm{RKG}}}(s) \\
&= \underset{1\le x\le M, 1\le y\le K}{\arg\min} \left\{ \mathbb{E}_n \left[\min_{1\le i\le M} \mathbb{E}[\max_{1\le j\le K} \theta_{i,j} | \mathcal{F}^{n+1}] \middle| S^n = s, (x^n, y^n) = (x,y) \right] - \min_{1\le i\le M} \mathbb{E}\left[\max_{1\le j\le K} \theta_{i,j} | S^n = s \right] \right\} \\
&= \underset{1\le x\le M, 1\le y\le K}{\arg\min} \left\{ \mathbb{E}_n \left[\min_{1\le i\le M} \mathbb{E}[\max_{1\le j\le K} \theta_{i,j} | \mathcal{F}^{n+1}] \middle| S^n = s, (x^n, y^n) = (x,y) \right] \right\}.
\end{aligned} \tag{11}$$

The second equality holds because $\min_{1\le i\le M} \mathbb{E}\left[\max_{1\le j\le K} \theta_{i,j} | S^n = s\right]$ is independent of (x,y).

We now have constructed a Doob martingale $\{\mathbb{E}[\max_{1\le j\le K} \theta_{i,j} | \mathcal{F}^n]\}_{n=1}^\infty$. However, the law that governs the evolution of $\{\mathbb{E}[\max_{1\le j\le K} \theta_{i,j} | \mathcal{F}^n]\}_{n=1}^\infty$ has no closed form, namely, the decision function $\mathcal{A}^{\pi^{RKG}}(\cdot)$ has no closed form. Regardless the closed form issue, we can still show that RKG is convergent and asymptotically optimal with the help of some mathematical analysis tools. In

application, we have to use Monte Carlo (MC) method to estimate the decision function. Even so, we can show that the convergence and asymptotic optimality are not affected by the perturbation of MC. Moreover, when $K=1$, the decision function reduces to KG and all the properties of RKG reduce to those of KG.

We can further obverse, via Jensen's inequality, that the value:

$$\mathbb{E}_n\left[\min_{1\le i\le M}\mathbb{E}[\max_{1\le j\le K}\theta_{i,j}|\mathcal{F}^{n+1}]\Big|S^n=s,(x^n,y^n)=(x,y)\right]-\min_{1\le i\le M}\mathbb{E}\left[\max_{1\le j\le K}\theta_{i,j}|S^n=s\right]$$

is always negative for any choice of (x,y) and is negatively correlated to $Var(\mathbb{E}[\max_{1\le j\le K}\theta_{i,j}|\mathcal{F}^{n+1}])$. This gives an intuition that no attractor exists if RKG is applied.

4.1. Optimality, Suboptimality and Convergence Results

We first assume that we can exactly compute the decision function of RKG. Then the RKG policy exhibits several optimality, sub-optimality and convergence properties. We only present theories here and proofs are left in appendix.

Firstly, any convergent policy is asymptotically optimal with respect to the objective function (6) . Secondly, RKG is a one-step optimal policy as shown in the definition of $\mathcal{A}^{\pi^{\text{RKG}}}(\cdot)$ and it is also a convergent policy. So RKG is asymptotically optimal. Thirdly, we can provide a bound on the suboptimal gap of RKG. All the results mentioned above are extension of optimality results proved in Frazier et al. (2009) for the KG policy. Comparing with KG, the objective function of RKG has one more layer. When $K=1$ the objective function becomes the same as the one of KG and all of the optimality results reduce to those of KG; when $K>1$, thanks to the nice properties of Gaussian distribution and the Lipschitz condition of $\max\{.\}$, all the properties of KG remain for RKG.

The following proposition shows that any convergent policy is asymptotically optimal.

Proposition 1. *Let our objective function be* $\min_{\pi\in\Pi}\mathbb{E}^{\pi}\left[\min_{1\le i\le M}\mathbb{E}[\max_{1\le j\le K}\theta_{i,j}|\mathcal{F}^N]\right]$. *Let* π *be a convergent policy. Then we have:* $\lim_{N\to\infty}V^0(s;N)=\lim_{N\to\infty}V^{0,\pi}(s;N)$ *for any* $s\in\mathbb{S}$.

We refer to this property as asymptotic optimality for it shows that the value function of a policy π converges to the optimal value function as number of sampling allowed goes to infinity. Proposition 1 is a direct result of benefits of measurement, which simply says that the objective function can be further minimized in expectation if more measurements are allowed. To show RKG get benefits from measurement, we need a slightly more sophisticated calculation than those in Frazier et al. (2008) for our objective function has one more layer with opposite operation. The proof is left in appendix.

Then the following simple proposition gives the result.

PROPOSITION 2. *RKG is a convergent sampling policy.*

The logics of proposition 2 is as follows. Suppose a set A of systems that RKG only takes finite number of samples exists. Let K be the number that $\pi^{RKG} \notin A$ after K times of sampling. As the state S^n converges to S^∞, benefits of measurement vanish on A^C the complement of A because $V^N(S^\infty) = \min_{x\in A^C} Q^{N-1}(S^\infty; y)$. Therefore, since any element in A can provide positive improvement, there must be another sampling decision on A after K times of sampling before we reach ∞. Proof of theorem 3 is similar to theorem 4 in Frazier et al. (2009). However, the preliminary of the proof is more relied on mathematical analysis compared with former researches of KG for we know little about the distribution of $min_{1\le i\le M}\mathbb{E}[\max_{1\le j\le K}\theta_{i,j}|\mathcal{F}^N]$.

Then, from proposition 1 and proposition 2 we can directly derive that:

THEOREM 3. *RKG is asymptotically optimal with respect to objective function*

$$\min_{\pi\in\Pi}\mathbb{E}^\pi\left[\min_{1\le i\le M}\mathbb{E}[\max_{1\le j\le K}\theta_{i,j}|\mathcal{F}^N]\right].$$

Now we know RKG is asymptotically optimal but asymptotically optimal polices have different rates of convergence. Asymptotic optimality is not equivalent to asymptotic rate of convergence. Convergence of a policy on its own indicates little about efficiency of the policy in the finite sample case. For instance, the equal allocation policy which allocates the computational budget in a round-robin fashion equally among the systems guarantees that every system is sampled infinitely often if infinite computational budget is available, and thus it is convergent. But its performance in the finite sample case is not particularly satisfying. Nevertheless, convergence should be a desired feature of a good sampling policy as it ensures that the policy does not "stick" in a proper subset of the systems, in which case the other systems would not be sampled infinitely often and thus would never be learned perfectly even given infinite computational budget. Proposition 1 is essentially a convergent result. It simply states that any convergent policy and the optimal policy can achieve the same asymptotic value by removing all the uncertainty to choose the correct underlying alternative. That is to say, our posterior knowledge about the correct alternative converges to perfect knowledge.

The third optimality result, which provides a general bound on suboptimality in the cases $1 < N < \infty$ not covered by the first two optimality results, is given by the following theorem. This bound is tight for small N and loosens as N increases. When $K = 1$, which means that the input distribution is known, and all the alternatives are independent, this bound is equal to the suboptimal bound of KG. We denote $||\tilde{\sigma}(\Sigma,\cdot,\cdot)|| := \max_{x,y,i}\tilde{\sigma}_i(\Sigma,x,y) + \min_{x,y,j}\tilde{\sigma}_j(\Sigma,x,y)$ and $||\Sigma|| := \max_i \Sigma_{ii}$.

THEOREM 4.

$$V^{n,\pi^{RKG}}(S^n) - V^n(S^n) \le \max_{(x^n,y^n),\ldots,(x^{N-2},y^{N-2})}\sum_{t=n+1}^{N-1}\sqrt{||\Sigma^k_{x^t:,x^t:}||2\log K} + \sqrt{2\pi}^{-1}||\tilde{\sigma}(\Sigma^t,\cdot,\cdot)|| \quad (12)$$

A proof of this theorem is given in appendix. Equation (12) is not a surprising result for its similarity to the bound of KG. The term $\sqrt{||\Sigma^k_{x^k:,x^k:}||2\log K}$ can be viewed as cost led by the inner opposite operation max{.}.

4.2. Optimality on Monte Carlo Estimate

Unfortunately, the decision function (11) has no closed form. We need to use MC to estimate its value. The MC estimator of RKG shares many similar properties with other stochastic optimization algorithms. For example, stochastic EM algorithm and stochastic gradient descent both converge to desired optimizer even if randomness occurs in the process of optimization. Here, we can show that MC estimated RKG converges to perfect information. Furthermore, MC estimator of any convergent policy is also convergent. Regarding to the estimation error, if the MC estimator gives a sampling decision other than the true one, since RKG is one-step optimal, we count the amount of improvement reduced as cost. We can show that the expected total amount of cost given any sampling budget N is bounded so the total cost can be arbitrarily small if we ensure that the estimator is precise enough. Therefore, the total error of MC estimated RKG is controllable.

We formally state several optimality result here: first, the random perturbation induced by MC does not affect the convergence and asymptotic optimality properties; second, the sub-optimal bound equals to the original one plus a perturbation of small order. We leave all the proofs in appendix as before and only state and briefly discuss these properties here.

The intuition that MC estimator of convergent policy is convergent is straightforward. We can see the sampling order as a Markov process on the set of system indexes. The convergence of a policy is equivalent to recurrence of its associated Markov process. Since RKG is convergent, its associated Markov process is recurrent on any index. If the Markov process is perturbed by a sequence of random variable with increasingly less randomness, it will still be recurrent on any index.

More precisely speaking, let π be a stationary convergent policy and $\mathcal{A}^\pi(\cdot)$ be its decision function and let $\hat{\mathcal{A}}^\pi(\cdot)$ be its MC estimator and $\hat{\pi}$ be the policy adopting $\hat{\mathcal{A}}^\pi$. We note that both $\hat{\mathcal{A}}^\pi$ and $\hat{\pi}$ are random. We let $p^\pi = (p^\pi_1, \ldots, p^\pi_{N-1}, \ldots)$ denote the sampling order $\Big((x^0, y^0), \ldots, (x^{N-1}, y^{N-1}), \ldots\Big)$ induced by π. We can view p^π as a recurrent Markov process on set $\{1, \ldots, M\} \times \{1, \ldots, K\}$ for the following reasons:

- For any $S^0 \in \mathbb{S}$, sampling results $\{\hat{z}^i\}_{i=0}^{N-1}$ are random.
- π selects each (i, j) infinitely often if infinite simulation budget is provided.

As n increases, p^π_n becomes more close to deterministic since the variance of θ is decreasing. For the same reason, $\hat{\mathcal{A}}^\pi$ becomes more accurate. So $p^{\hat{\pi}}_n(s^n)$ is close to $p^\pi_n(s^n)$ for any $s^n \in \mathbb{S}$. If $\hat{\pi}$ is not a convergent policy, then $p^{\hat{\pi}}$ is transient on set $T \subset \{1, \ldots, M\} \times \{1, \ldots, K\}$. The transience implies

that MC gives wrong estimation on systems in T with probability 1 even if we have increasingly more accurate MC estimation. This event is of 0 probability measure if $\mathbb{P}(\hat{\pi}=\pi)>0$. So we have the following theorem.

THEOREM 5. *Let $\hat{\pi}^{RKG}$ a consistent MC estimator of π^{RKG}. $\hat{\pi}^{RKG}$ is convergent almost surely.*

From this theorem and proposition 1, we can easily derive the following theorems.

THEOREM 6. *Let $\hat{\pi}^{RKG}$ be a consistent MC estimator of π^{RKG}. $\hat{\pi}^{RKG}$ is asymptotically optimal.*

The last result is about the sub-optimal bound of $\hat{\pi}^{RKG}$ in the case $1<N<\infty$. Let L be the number of samples generated by MC. Then the MC estimator converges to the true value with rate $O(\frac{1}{\sqrt{L}})$. We first define the cost that MC makes a wrong decision:

$$C^n(s)=\sum_{(x,y)}\mathbb{P}(\mathcal{A}^{\hat{\pi}^{RKG}}(s)=(x,y))\big[\mathbb{E}[\min_i\mathbb{E}[\max_j\theta_{ij}|S^{n+1}]\big|S^n=s,(x^n,y^n)=(x,y)] \\ -\min_{(x',y')}\mathbb{E}[\min_i\mathbb{E}[\max_j\theta_{ij}|S^{n+1}]\big|S^n=s,(x^n,y^n)=(x',y')]\big]$$

which is finite. Then when N is finite, the total expected cost $\sum_{n=1}^{N}C^n(S^n)$ is still finite almost surely. Some people may wonder what if the sampling budget is infinite and in that case, even if we can control the error cost in each sampling by choosing L large enough, the infinite-series sum $\sum_{n=1}^{\infty}C^n(S^n)$ may blow up to infinity. This will never happen in our setting since the total improvement is finite and the total cost must not be greater than the total improvement for the reason that benefits of measurement tells us that any choice of measurement gives positive improvement and if the total cost is greater than the total improvement then some measurements give negative improvement, violating the property.

We should also notice that MC is making increasingly more accurate decision because the randomness of systems decreases after each sampling. In fact, since the total improvement by any policy π and given any budget N is bounded by a constant in expectation as we will show in the appendix, we can have the following inequality about the total cost:

$$\begin{aligned}
&\mathbb{E}\sum_{n=0}^{\infty}C^n(S^n)=\mathbb{E}\sum_{n=0}^{\infty}|V^{N-1,\ \hat{\pi}^{RKG}}(S^n)-V^{N-1,\ \pi^{RKG}}(S^n)|\\
&=\mathbb{E}\sum_{n=0}^{\infty}V^{N-1,\ \hat{\pi}^{RKG}}(S^n)-V^{N-1,\ \pi^{RKG}}(S^n)\\
&\le\mathbb{E}\sum_{n=0}^{\infty}V^{N,\pi^{RKG}}(S^n)-V^{N-1,\pi^{RKG}}(S^n)\\
&=\sum_{n=0}^{\infty}\mathbb{E}[V^{N,\pi^{RKG}}(S^n)-V^{N-1,\pi^{RKG}}(S^n)]\\
&\le V^{N,\pi^{RKG}}(S^0)-U(S^0)\\
&<\infty
\end{aligned}$$

where the second line is because π^{RKG} is a one-step optimal policy; the third line is from benefits of measurement; the fourth line is from Tonelli's theorem; on the fifth line, $U(S^0)$ is the lower bound of value function of any policy π given initial state S^0. Therefore, the total cost from wrong decisions is finite. Moreover, since the value function difference between $\hat{\pi}^{RKG}$ and π^{RKG} converges to zero in expectation as $L \to \infty$ and the difference is always positive, We can further derive that:

$$\lim_{L\to\infty} \sum_{n=0}^{\infty} C^n(S^n) = 0 \qquad \text{in } L^1$$

From the above discussion, the following theorems are very intuitive:

THEOREM 7.

$$V^{n,\hat{\pi}^{RKG}}(S^n) - V^n(S^n) \le \frac{C}{\sqrt{L}} + \max_{(x^n,y^n),\ldots,(x^{N-2},y^{N-2})} \sum_{t=n+1}^{N-1} \sqrt{2||\Sigma^k_{x^t:,x^t:}||\log K} + \sqrt{2\pi}^{-1}||\tilde{\sigma}(\Sigma^t,\cdot,\cdot)||. \tag{13}$$

where C is some constant independent of n and $\frac{C}{\sqrt{L}}$ converges to 0 as $L \to \infty$. We will prove this theorem in appendix.

THEOREM 8. *Given any $S^0 \in \mathbb{S}$, $\sum_{n=0}^{\infty} C^n(S^n) \to 0$ in L^1 as L, the number of samples drawn in each MC estimation, tends to ∞.*

Proof:

Because the MC estimator of RKG is consistent, it converges to the true value almost surely as $L \to \infty$ from the Strong Law of Large Number. So we have:

$$\lim_{L\to\infty} \mathbb{P}\{\mathcal{A}^{\hat{\pi}^{RKG}}(s) \ne \mathcal{A}^{\pi^{RKG}}(s)\} = 0$$
$$\lim_{L\to\infty} \mathbb{P}\{\mathcal{A}^{\hat{\pi}^{RKG}}(s) = \mathcal{A}^{\pi^{RKG}}(s)\} = 1$$

for any $s \in \mathbb{S}$ and hence $\lim_{L\to\infty} C^n(s) = 0$ for any $s \in \mathbb{S}$.

$C(S^n) \ge 0$ is bounded from above by $\min_i \mathbb{E}[\max_j \theta_{ij}|S^n] - \mathbb{E}\big[\min_i \mathbb{E}[\max_j \theta_{ij}|S^{n+1}\big|S^n,(x^n,y^n) = \mathcal{A}^{\pi^{RKG}}(S^n)] \ge 0$ which is in L^1 according to content in appendix and definition of $C^n(.)$. So by dominated convergence theorem, we have:

$$\lim_{L\to\infty} \mathbb{E}[C^n(S^n)] = \lim_{L\to\infty} \int_{\mathbb{S}} C^n(s) d\mathbb{P}^{\hat{\pi}^{RKG}}_{S^n}(s) = \int_{\mathbb{S}} \lim_{L\to\infty} C(s) d\mathbb{P}^{\hat{\pi}^{RKG}}_{S^n}(s) = 0$$

where $\mathbb{P}^{\hat{\pi}^{RKG}}_{S^n}(.)$ is the probability measure of S^n induced by $\hat{\pi}^{RKG}$. On the other hand, from the previous content, we have $\mathbb{E}\sum_{n=1}^{\infty} C^n(S^n) < \infty$. Given any $\epsilon > 0$, we can find a number N such that $\mathbb{E}\sum_{n=N}^{\infty} C^n(S^n) = \sum_{n=N}^{\infty} \mathbb{E}C^n(S^n) < \frac{\epsilon}{2}$. Then for $\mathbb{E}\sum_{n=N}^{N} C^n(S^n)$ we can choose L large enough such that $\mathbb{E}\sum_{n=N}^{N} C^n(S^n) < \frac{\epsilon}{2}$. Therefore, $\mathbb{E}\sum_{n=1}^{\infty} C^n(S^n) < \epsilon$ for L large enough. Since $C^n(s) \ge 0$ for any $s \in \mathbb{S}$, we finally can derive that $S^0 \in \mathbb{S}$, $\sum_{n=0}^{\infty} C^n(S^n) \to 0$ in L^1 as L tends to ∞.

□

5. NUMERICAL EXPERIMENTS

We compare performances of policies with several numerical experiments in this section. We first present a numerical experiment with standard Bayesian optimization problem framework in section 6.1; and then we present application to the revenue of production line in section 6.2 and the (s, S) order policy in section 6.3.

We have introduced two stationary policies for sequential sampling of the Bayesian robust R&S problem, i.e. NKG and RKG. We will also include three additional polices as follows in the numerical experiments.

- *Equal allocation* (EA). The sampling decisions are determined in a round-robin fashion: the sequence of decisions are $(1,1),(2,1),\ldots,(M,1),(1,2),(2,2),\ldots,(M,2),\ldots,$ $(1,K),(2,K),\ldots,(M,K)$ and repeat the sequence if necessary.
- *Maximum variance* (MV). The sampling decision at each time n is to choose system (i,j) that has the maximum variance $\Sigma^n_{(i,j),(i,j)}$.
- *Maximum adaptively weighted knowledge gradient (MAWKG). The value* $\mathbb{E}_n\left[\max_j \mu^{n+1}_{x,j}\middle| S^n = s, (x^n, y^n) = (x,y)\right] - \max_j \mu^n_{x,j}$ *is defined as the uncertainty of system* (x,y)*. Let* w^n_i *be an estimate of* $P(i = \arg\min_k\{max_m \theta_{k,m}\}\big|\mathcal{F}^n)$*. MAWKG adaptively selects the system with the greatest weighted uncertainty:*

$$\mathcal{A}^{\pi^{\text{MAWKG}}}(s) = \underset{1\le x\le M, 1\le y\le K}{\arg\max}\, w^n_x \left\{\mathbb{E}_n\left[\max_{1\le j\le K}\mu^{n+1}_{x,j}\middle| S^n = s, (x^n,y^n) = (x,y)\right] - \max_{1\le j\le K}\mu^n_{x,j}\right\}, s\in\mathbb{S}.$$

For discussion of MAWKG in detail, please refer to Zhang and Ding (2016).

5.1. Standard Example Problem

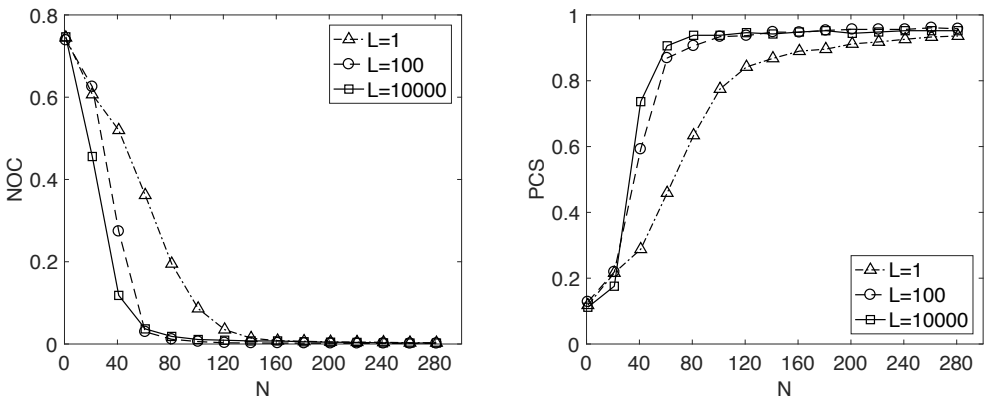
We first compare the performances of RKG policy with different L where L is the number of samples drawn in MC and then we compare different policies. All of our comparisons are under a standard Bayesian framework. The comparison is based on 1000 randomly generated problems, each of which is parameterized by a number of sampling opportunities N, a number of systems $M\times K$, an initial mean $\mu^0\in\mathbb{R}^{M\times K}$, an initial covariance matrix $\Sigma^0\in\mathbb{R}^{MN\times MN}$, and sampling variance $\delta^2_{i,j}$, $i=1,\ldots,M$, $j=1,\ldots,K$. Specifically, we set $M=K=10$ and $\delta_{i,j}=1$ for each (i,j), and choose Σ^0 from the class of power exponential covariance functions, particularly

$$\Sigma^0_{(i,j),(i',j')} = \begin{cases} 100e^{-|j-j'|^2}, & \text{if } i=i', \\ 0, & \text{if } i\neq i'. \end{cases}$$

Each $\mu^0_{i,j}$ is generated independently according to the uniform distribution on $[-1,1]$.

For each randomly generated problem, the true value θ is generated according to the prior belief of the problem, i.e. $\mathcal{N}(\mu^0,\Sigma^0)$. In the motivational robust R&S problem (1), we interpret M as

Figure 1 NOC and PCS based on 1000 randomly generated problems.



the number of possible decisions or alternatives s_i of a simulation model, and K as the number of possible input distributions P_j. We argue that a decision-maker relying on the simulation model is more concerned of the decision s_i than of the distribution P_i. Suppose that we select system (x^N, y^N) at time N, i.e. $\mu^N_{x^N,y^N} = \min_i \max_j \mu^N_{i,j}$. Let system (i^*, j^*) be the true optimal system, i.e. $\theta_{i^*,j^*} = \min_i \max_j \theta_{i,j}$. Then, we consider it as a correct selection if $x^N = i^*$, regardless of the value of l. In other words, it really matters to select the correct alternative and not so much with the correct input distribution. In addition to the probability of correct selection in the above sense, we also compare policies based on normalized opportunity cost (NOC) of incorrect selection

$$\frac{\left|\theta_{i^*,j^*} - \max_j \theta_{x^N,j}\right|}{\sqrt{\frac{1}{MK}\sum_{i,j} |\theta_{i^*,j^*} - \theta_{i,j}|^2}}. \tag{14}$$

We apply all the completing policies on the 1000 randomly generated problem for different values of N to observe how each policy converges. For a fixed N, we record for each problem whether a policy selects the correct alternative after N sampling decisions as well as the realized NOC (14). By doing so, we estimate probability of selecting the correct alternative and NOC for each policy given N.

The following experiments presents various statistics of the realized NOC and PCS for representative values of N. Note that each problem consists of $MK = 100$ systems. Hence, $N = 100$ represents a scenario where one has sufficient computational budget whereas $N = 50$ and $N = 20$ represent normal and low budgets, respectively.

Figure 1 indicates how NOC and PCS change with L. We define policy π_1 as the MC estimated RKG with $L = 1$ and similarly, we define π_2 and π_3 for the case $L = 100$ and $L = 10000$ respectively. We note that π_1 has the worst performance and as the sampling budget N increases from 20 to 300, PCS and NOC of all policies converge to the same levels. This is because the MC estimation

of π_1 is very rough in early stages. When L is small and $|\Sigma^n|$ is relatively large in these early stages, the MC estimation has high variance and hence the selection by RKG may largely deviate from the correct one. So, the decision by π_1 is approximately uniformly distributed. In this sense π_1 is close to equal allocation. As the sampling budget N becomes larger, $|\Sigma^n|$ becomes smaller with more sampling opportunity. When $|\Sigma^n|$ is small enough, we can make precise MC estimation even if $L=1$. As a result, π_1 converges to π_2 and π_3 in NOC and PCS.

From equation (13), we know that MC estimator of RKG converges to the true one in $\frac{1}{\sqrt{L}}$. So we can say that π_{i+1} is ten times better than π_i for $i=1$, 2. When $L=100$, the precision of MC estimation is high compared with the case $L=1$. In contrast, the improvement from $L=100$ to $L=10000$ is not that obvious. This is because we only need to make decision on a finite discrete set $\{1,...,M\}\times\{1,...,K\}$ and the gaps between values

$$\{\mathbb{E}\big[\min_i \mathbb{E}[\max_j \theta_{i,j}|S^{n+1}]|S^n,(x^n=x,y^n=y)\big]\}_{(x,y)\in\{1,...,M\}\times\{1,...,K\}}$$

relax the precision requirement of MC. We deem that when $L=10000$ the policy has the best performance in balancing PCS and the number of L. In the following experiments, we set $L=10000$ for RKG.

Now we compare RKG with other policies. Table 1 indicates the NOC of five different policies. We note that RKG has the smallest expected NOC throughout.

Our numerical experiments show that the relative performance of the five competing policies changes considerably for different levels of computational budget. The best policy is different depending on if the budget is low, normal, or sufficient and on how we define the performance of a policy. First, if the computational budget is low, RKG has the lowest PCS and NOC; surprisingly, the policy with the lowest NOC in early stages is not the one with the highest PCS. We believe that this is because at the beginning stage of sequential sampling, our prior knowledge are significantly different from the true value of θ, so information are too noisy and thus our effort of exploration is severely misled. However, RKG successfully rules out systems which is unlikely to be our target and narrows down the set that needs to take more samples on, leading to low NOC. In other words, it needs certain "warm-up" stage for performance improvement. This "warm-up" stage is due to low correct rate of target row guessing in the first few rounds. Intuitively, RKG first focus on the potential target row based on current knowledge. It is a trade-off between guessing and knowledge. High amount of knowledge is awarded to correct guess. In contrast, wrong guess is penalized by below-average amount of knowledge. That is the reason why it needs "warm-up" stage since it is hard to make a correct guess when knowledge about the system is rough. So the PCS in the early stages is low. Even so, wrong guess helps us rule out some missleading systems and hence highly reduce our distance to the true value.

Table 1 Opportunity cost of selecting an incorrect alternative based on 1000 randomly generated problems.

Budget	Stat.	Sampling Policy				
		EA	MV	NKG	MAWKG	RKG
$N = 20$	Q_1	0.2145	0.1535	0.1778	0.1795	0.1801
	Median	0.6276	0.5569	0.4137	0.6874	0.6944
	Q_3	1.0273	0.9421	0.8306	1.1145	1.2279
	Max	2.6338	2.6750	3.3601	2.2792	2.3065
	Mean	0.6842	0.6020	0.5693	0.7092	$\boxed{0.4699}$
$N = 50$	Q_1	0.0000	0.0000	0.0000	0.0000	0.0000
	Median	0.3384	0.0407	0.0088	0.0000	0.0000
	Q_3	0.8074	0.4849	0.3788	0.0000	0.0000
	Max	2.1869	2.1459	2.4811	1.7004	1.2654
	Mean	0.4755	0.3022	0.2669	0.0836	$\boxed{0.0798}$
$N = 100$	Q_1	0.0000	0.0000	0.0000	0.0000	0.0000
	Median	0.0000	0.0000	0.0000	0.0000	0.0000
	Q_3	0.0000	0.0000	0.3476	0.0000	0.0000
	Max	2.0913	0.4792	2.9566	0.4949	0.4523
	Mean	0.0325	0.0149	0.2598	0.0128	$\boxed{0.0085}$

The boxed numbers indicate the smallest means among all the policies. Q_1 and Q_3 denote the first and third quartiles, respectively.

As more computational budget is available, the performance of RKG improves dramatically which shows the power of correct guess and ruling out missleading systems. In particular, with normal computational budget, RKG also produces the smallest NOC and it is significantly better than the second best policy MAWKG in term of PCS; the worst performance is delivered by EA, which is not surprising since it utilizes no information of the systems at all.

At last, if the computational budget is sufficiently high, then all the policies except NKG produce small NOC, which implies that they are able to identify the optimal system, or at least the optimal row of θ, with sufficiently many sampling opportunities. The only exception, NKG, fails to do so and its NOC is at least one order of magnitude larger than the others.

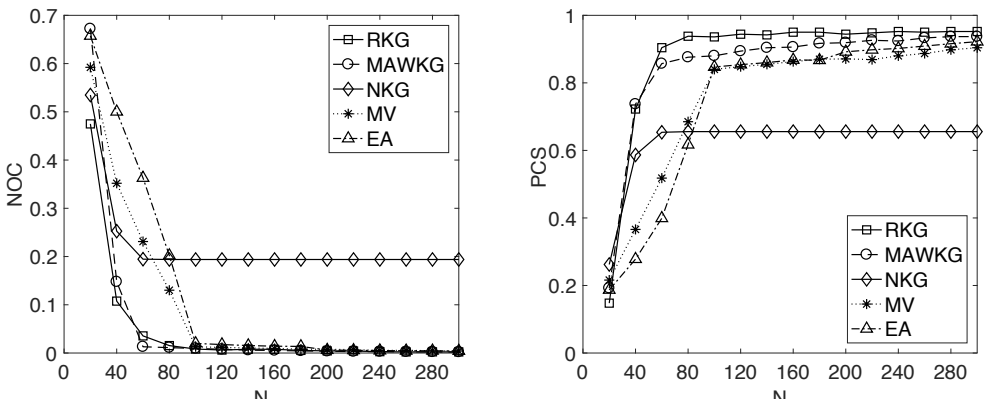
The PCS plot in figure 2 illustrates the asymptotic behavior of the seven competing policies in terms of probability of selecting the correct alternative as the computational budget increases. The conclusions we draw from Figure 2 are consistent with those from Table 1. First, all the policies except NKG are convergent. Second, the RKG obviously outperforms other policies in the normal and high budget cases.

In the standard Bayesian framework, MAWKG and RKG obviously outperform other policy. In the following real application example problems, we will use MAWKG as a benchmark of best-case performance. In contrast, we use EA or MV as the worst case performance benchmark.

5.2. Production Line Management

We consider the following revenue maximization problem based on Buchholz and Thummler (2005) and "Optimization of a Production Line" from the testbed of SimOpt(Pasupathy and Henderson

Figure 2 NOC and PCS based on 1000 randomly generated problems.



2006). A factory has a production line with M workstations arranged in a row. Each workstation follows a first-come-first-serve discipline. Parts leaving queue of workstation n after service are immediately transferred to the next queue of workstation $n+1$. Whenever the work load of workstation $n+1$ is full, the workstation n is said to be blocked and the part in it cannot leave even if it is completed, since there is no room in the next queue. The capacity of each workstation is finite and equal to K. Parts arrive to the production line according to a Poisson process with rate λ.

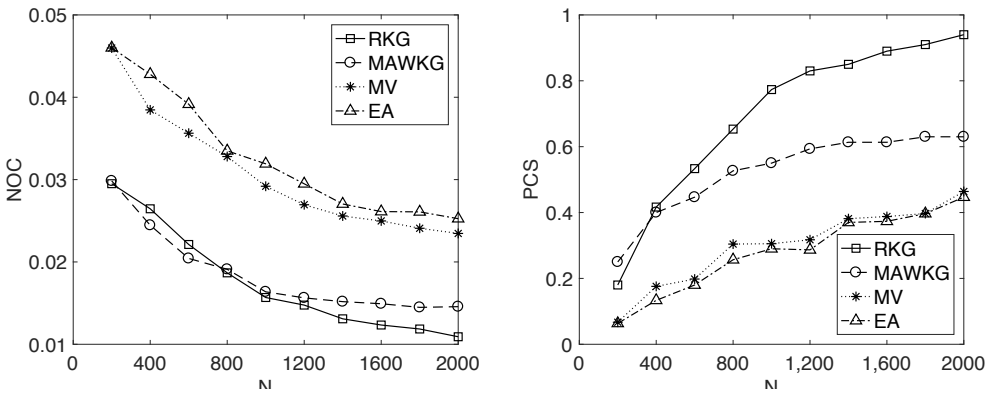
The service time of each workstation is exponentially distributed but the service rate s_n is unknown. The vector of cost for running workstations is denoted as $\overrightarrow{c}$. Suppose that the plant manager has prior knowledge of the service rate, so she can restrict the rate in a set finite $\mathbb{U}$. Given a time horizon t, the throughput of the production line is defined as the average number of parts leaving the last queue in unit time, denoted $W = W(\overrightarrow{s})$, where $\overrightarrow{s} = (s_1, \ldots s_N)$. Assume the decision variable is the vector of capacity $\overrightarrow{K}$. The objective is then to choose a vector of capacity that maximizes the revenue function over worst case:

$$\max_{k\in\mathbb{Z}} \min_{u\in\mathbb{U}} \mathbb{E}\left[\frac{W(\overrightarrow{\mu})}{1+\overrightarrow{c}\cdot\overrightarrow{s}}\middle| K=k, \overrightarrow{s}=u\right]. \tag{15}$$

We assume the production line has 3 workstations all of which have an equal capacity K and an equal service rate μ. Both K and μ are unknown but we assume that $K \in \{6, 7, \ldots, 15\}$ and $s \in \{0.4, 0.5, \ldots 1.3\}$. We set the arrival rate of parts $\lambda = 1$. The time length of running each simulation is 1000×unit time. Then the true underlying value $\theta_{i,j} := \mathbb{E}\left[\frac{W(\overrightarrow{\mu})}{1+\overrightarrow{c}\cdot\overrightarrow{s}}\middle| K=i, s=j\right]$; each sampling on system (i,j) is a simulation of the production line with capacity $K=i$ and service rate $s=j$. We also assume that all the systems are independent to each other.

We can run a large number of simulations to approximate the revenue function on each pair $(s,u) \in \mathbb{Z}\times\mathbb{U}$ and choose the one that satisfies equation (15). Obviously, this is not an efficient way. In the Bayesian framework, we can assume that $\theta_{i,j}$ are normally distributed and if we have prior knowledge μ^0 and Σ^0 of θ, then we can use RKG or MAWKG to update our prior and

Figure 3 NOC and PCS based on 300 randomly generated initial μ^0.



thus we can select the argument that satisfies the max min problem after simulation budget is exhausted. Even better, we do not need to know μ^0 so we can assign random value to μ^0. In the experiment, we randomly generate independent μ^0_{ij}'s from the uniform distribution on $[-1, 1]$. We use the common sampling precision in Xie and Frazier (2013) to determine the sampling error. The sampling precision equals to the inverse of covariance matrix in the independent sampling case and the update rule for prior precision is fully discussed in Frazier et al. (2008). More precisely, we randomly chose 5 systems on each row, sampled 20 times from each of them to estimate their individual sampling precisions, and used the average of the 5 sampling precisions as the estimate of the common sampling precision for each row. All the estimated precision is in the order of 10^{-3}. We use independent normal prior for each system. We assign a randomly generated number to each prior mean, and the common sampling precision to each prior precision. This is equivalent to using a non-informative prior and starting sampling by taking a single sample from each alternative.

We then take 10000 sampling on each (i, j) to estimate θ and select the objective decision variable i^* from these estimators. We then run RKG, MAWKG, MV and EA with simulation budget $N = 200 : 200 : 2000$. After the budget is exhausted,depending on the applied sampling policy π, we make a decision i^π and compare it with i^*. If $i^\pi = i^*$, we count it as correct selection. We apply the competing policies to 300 randomly generated μ^0 for different values of N and take the ratio of correct selection as PCS.

Figure 3 shows the comparison results. As what we have seen in the previous numerical experiment, RKG is not the best choice when the simulation budget N is low but it dominates other policies as N increases. Performance of the second best policy MAWKG is not even close to RKG under normal and high simulation budget scenarios. MV and EA have the worst performance. In fact, EA and MV are nearly the same except in few early stages because when $\Sigma^n_{(i,i),(j,j)} \approx \Sigma^n_{(k,k),(l,l)}$, $\forall i, j, k, l$, MV allocate sampling effort evenly to all systems. As a result, MV and EA have close PCS's.

5.3. (s, S) Policy

We apply the RKG, MAWKG, MV and EA to analyze different order strategies for an inventory system. The example problem is adapted from Kleijnen et al. (2010) and the testbed of SimOpt (Pasupathy and Henderson 2006). We consider a (s, S) inventory model with full backlogging. Demand during each period D_t is exponential distributed with unknown mean γ. Our manager knows that the true γ belongs to a finite set Γ from historical data and statistics. The inventory position IP_t, which is equal to on-hand inventory-backorders+orders, of period t is calculated at the end of period t. If $IP_t \leq s$, then we make a replenishment order with quantity $S - s$ to get back up to S. We assume that lead times are Poisson distributed with mean λ and all replenishment orders are received at the beginning of the period. Note that an order with lead time l placed in period t will arrive at the beginning of period t+l+1 for we place the order at the end of period t. Let $h = 1$ be the unit holding cost for inventory on-hand; furthermore, there is a fixed setup cost A and a variable, per unit, production cost c. our goal is to find a (s, S) order policy that can minimize the expectation of total cost C per period over worst case:

$$\min_{(s,S)\in\mathbb{Z}^2} \max_{\gamma\in\Gamma} \mathbb{E}\left[C\middle|(s,S),\gamma\right]. \tag{16}$$

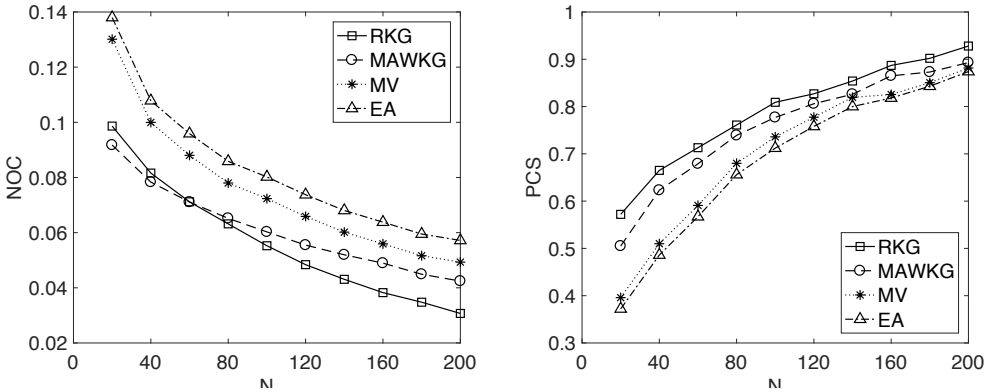
By following the suggested parameter setting and starting solutions, we let $A = 36$, $c = 2$, $\lambda = 6$ and let $\Gamma = \{50, 70, \ldots, 150\}$ and $(s, S) \in \mathbb{S} = \{(1000, 2000), (700, 1500), (1000, 1500), (700, 2000), (800, 1700), (100, 500)\}$. The time length of running each simulation is 60 periods, 50 days for warm-up periods and 10 days for simulation. The expected total cost for each $((s, S), \gamma)$ is the underlying values we want to know: $\theta_{(s,S),\gamma} := \mathbb{E}\left[C\middle|(s,S),\gamma\right]$. We assume all the systems are independent to each other.

As in the production line management, our setting can be extended to more general scenarios: the sets Γ and $\mathbb{S}$ could be larger and input uncertainty could be placed on lead time l. However, these will lead to a higher order of computational time complexity.

We run 10000 simulations on each system $((s, S), \gamma)$ and use the average to approximate $\theta_{(s,S),\gamma}$. Then we choose the optimal argument (s^*, S^*) that satisfies objective function (16). We view (s^*, S^*) as the true objective decision variable in the dynamic programming problem.

As before, we assume systems are independent to each other. So we use common sampling precision to determine the precision of sampling. All the common sampling precisions are in the order of 10^{-2}. We then assign random values to μ^0_{ij}'s generated from the uniform distribution on $[-1, 1]$ as our prior mean and take sampling precisions as prior precision. For different simulation budget $N = 20 : 20 : 200$, we test the PCS of four competing policies, namely RKG, MAWK, EA and MV, on 1000 randomly generated μ^0 for each N. For a policy π, when simulation budget

Figure 4 NOC and PCS based on 1000 randomly generated initial μ^0.



is exhausted, we select $(s^\pi, S^\pi) = \arg\min_{(s,S)\in\mathbb{S}}\{\max_{\gamma\in\Gamma}\mu^N_{(s,S),\gamma}\}$. Similar to the production line management problem, if $(s^\pi, S^\pi) = (s^*, S^*)$ when the simulation budget is exhausted then we count it as correct selection and the PCS of π is defined as the ratio of correct selections.

Figure 4 shows the comparison results. It is similar to previous comparisons except that PCS's of all policies are higher than those in figure 3. It is obvious that the PCS curve of RKG has the most fastest rate converging to 1. MV and EA have approximately equal bad performances. Performance of MAWKG is in between.

To summarize, all the numerical experiments strongly demonstrate that RKG is an ideal sequential sampling policy given that the simulation budget is not too low or that our prior knowledge about the objective is not too rough. In early stages, it may make some wrong guessing of the objective alternative. However, there is a trade-off between wrong guessing and fast convergence rate. Once RKG approximately figures out the region of the objective alternative, it can concentrate on that region and reduce a large amount of uncertainty about the objective.

6. Conclusions

In this article, we consider the sequential sampling Bayesian R&S problem in the presence of input uncertainty and show that, depending on the objective function, not all one-step optimal Bayesian policy is convergent. In order to have a robust policy, we extend the KG policy proposed in Gupta and Miescke (1996) and Frazier et al. (2008) for Bayesian R&S problem which assume that the distribution of input is known.

We first prove the non-convergence property of a naive extension of KG; we then introduce another extension and present its asymptotic optimality and sub-optimality bound. We name the non-convergent policy as naive knowledge gradient(NKG) and the convergent one as robust knowledge gradient (RKG). Since the sampling decision function of RKG has no closed form, we use Monte Carlo method to estimate its value; further, we prove the convergence and asymptotic

optimality are preserved on MC and the sub-optimality bound is perturbed by a value of order $\frac{1}{\sqrt{L}}$, where L is the number of MC sampling.

We show the robustness of RKG by different numerical experiments and it turns out that, except in the low simulation budget case, RKG is a highly efficient sequential sampling policy in applications with large number of alternatives and uncertain input distribution for which the way to achieve a robust solution is by balancing the concentration on potential alternatives and the removal of systems' randomness, and the sequential nature of RKG allows higher efficiency by concentrating later measurements on alternatives revealed by earlier measurements to be among the best; meanwhile, randomness of other systems are also considered by it. The defect of RKG is its low PCS in the low simulation budget case. This is caused by wrong information in the early stages. In other words, the low PCS is not caused by the nature of RKG but, instead, by incorrect prior knowledge.

We would also like to mention that RKG can be generalized to other Bayesian R&S problems when input uncertainty appears. Once a problem is formulated in the Bayesian framework, the future requirement for applying approach similar to RKG is to restrict the input uncertainty in a finite set and calculate, exactly or by approximation, the quantity $\arg\min_{(x,y)} \mathbb{E}^{\pi}[\min_i \mathbb{E}[\max_j \theta_{ij} \Big| \mathcal{F}^{n+1}] \Big| (x^n, y^n) = (x, y)]$ as shown before. For example, it can be used to multiple comparisons with a standard (MCS) problem (Xie and Frazier 2013) in the presence of input uncertainty: an adaptive stopping rule rather than a fixed sampling budget could be used; objective other than the expected cost of the selected alternative, such as deviation from a desired standard, could be considered. Moreover, RKG can also be applied to R&S problems with other types of uncertainty, depending on the actual situation. At last, we believe that, when facing R&S problem with input uncertainty, the approach of building a Bayesian framework, restricting the input uncertainty in a finite set and the calculating a RKG policy adapted to the actual problem promises acceptable results in a large number of real applications.

A.1. NKG Policy

We first give proof to theorem 1.

Proof of Theorem 1:

$1 \to 3$: This is a direct result of the law of large number.

$3 \to 1$: We choose the objective that π needs to identify the order set of θ. Then since no perfectly correlated systems exists, π must take sample on every system infinitely often in order to identify the true order set.

$1 \to 2$: Define an operation $U^{(x,y)}$ of covariance matrix Σ according to equation 2. From direct computation, we can have:

$$||U^{(1,1)}U^{(1,2)}\dots U^{(M,K)}\Sigma|| < ||\Sigma||$$

where $||\cdot||$ is the operator norm. From Lemma 2, we know that $U^{(x,y)}$ and $U^{(}x',y')$ commute for any (x,y) and (x',y'). Therefore, if policy π samples each system infinitely often, then we have:

$$\Sigma^{\infty} = \big[\lim_{n\to\infty}\prod_{k=1}^{n} U^{((x^k(\pi),y^k(\pi))}\big]\Sigma^0 = \big[\lim_{n\to\infty}\prod_{k=1}^{n} U^{(1,1)}U^{(1,2)}\dots U^{(M,K)}\big]\Sigma^0 = 0.$$

$2 \to 1$: We suppose that 1 is not true. Then exists $(x^*.y^*)$ and N such that $(x^k(\pi).y^k(\pi)) \neq (x^*,y^*)$ for any $k \geq N$. Then since $|\Sigma^0| > 0$, we can easily derive that $|\Sigma^N| > 0$ and that the set $\big\{\frac{\Sigma^N_{x:,x:}e_y e_y^T \Sigma^N_{x:,x:}}{\delta^2_{x,y}+\Sigma^N_{(x,y),(x,y)}} : (x,y) \in (1,\dots,M)\times(1,\dots,K)\big\}$ are linearly independent. Now we define:

$$V := \frac{\Sigma^N_{x^*:,x^*:}e_{y^*}e_{y^*}^T\Sigma^N_{x^*:,x^*:}}{||\Sigma^N_{x^*:,x^*:}e_{y^*}e_{y^*}^T\Sigma^N_{x^*:,x^*:}||}.$$

So the range of $\lim_{n\to\infty}\prod_{k=N+1}^{n} U^{((x^k(\pi),y^k(\pi))}$ is in $\mathcal{M} := \bigoplus_{(x,y)\neq(x^*,y^*)}\Sigma^N_{x:,x:}e_y e_y^T\Sigma^N_{x:,x:}$. We then define:

$$V^* = \frac{V - P_{\mathcal{M}}V}{||V - P_{\mathcal{M}}V||_2}$$

where $P_{\mathcal{M}}$ is the projection operator into the range of all the matrices in $\mathcal{M}$. So we have

$$||\big[\lim_{n\to\infty}\prod_{k=N+1}^{n} U^{((x^k(\pi),y^k(\pi))}\big]\Sigma^N V^*||_2 = ||\Sigma^N V^*|| > 0$$

so 2 is not true.

$4 \to 2$: 4 implies 2 directly.

1, $2 \to 4$: from law of large number, we know that if every system is sampled infinitely often then $\mu^n \to \theta$. Also, 2 tells that if 1 is satisfied, then $\Sigma^n \to 0$. So 1 and 2 imply 4. However, 1 and 2 are equivalent so either 1 or 2 implies 4.

□

The Objective function of NKG is defined as:

$$\mathbb{E}^{\pi}[\min_i \max_j \mu_{ij}^N] \tag{17}$$

where π is a policy in the policy space Π and N is the number of measurements allowed. So given a particular state $s=(\mu,\Sigma)\in\mathbb{S}$ in the nth measurement, NKG chooses the measurement position (i^*,j^*) which satisfies:

$$(i^*,j^*)=\arg\min_{(i,j)} \mathbb{E}[\min_i \max_j \mu_{ij}^{n+1}|S^n=s]. \tag{18}$$

According to our numerical simulation, we found that NKG policy is not convergent. Before we give the formal proof, we would like to give the general idea of the proof. We first simplify the notation. Equation (17) is equivalent to the following problem:

$$\begin{aligned}
&\mathbb{E}[\min_D \sum_i D_i \max_j \mu_{ij}^{n+1}|S^n=s]\\
=&\mathbb{E}[\min_D \sum_i D_i(\max_j \mu_{ij}^n+\max_j \mu_{ij}^{n+1}-\max_j \mu_{ij}^n)|S^n=s]\\
&s.t.\quad D\geq 0\\
&\sum_i D_i=1.
\end{aligned}$$

We then define:

$$\begin{aligned}
V_i^n &:= \max_j \mu_{ij}^n\\
O_i^n &:= \max_j \mu_{ij}^{n+1}-\max_j \mu_{ij}^n.\\
\delta_i(x) &= \begin{cases} 0, & \text{if } x\neq i\\ 1, & \text{otherwise}\end{cases}
\end{aligned}$$

When $S^n=s$ is known, we have V^n is deterministic and O^n is a random variable whose distribution depends on measurement position (x,y). Suppose measurement position is (x,y), for all $i\neq x$ we have $O_i^n=0$ for each row of $\mu_i j$ is independent. Furthermore, even if $O_i^n\neq 0$, it also depends on y. So we can write $O_i^n=O_i^n(y)$ in this case. Therefore, we further simplify (17) as:

$$\begin{aligned}
&\min_{(x,y)} \mathbb{E}^n\{\min_D \sum_i D_i[V_i^n+\delta_i(x)O_i^n(y)]\}\\
&s.t.\ D\geq 0\\
&\sum_i D_i=1.
\end{aligned} \tag{19}$$

Now equation (19) can be considered as a linear programming problem if O^n is deterministic. Suppose O^n is small, deterministic and non-negative then if we choose $x=\arg\min_i V_i^n$, equation (19) is larger than $\min_i V_i^n$ and if we choose $x\neq\arg\min_i V_i^n$, the value of (19) does not change. Therefore, $x=\arg\min_i V_i^n$ is ruled out.

In our case, O^n is random. However it is positive in expectation and becomes closer to a deterministic vector as n becomes larger which means more samplings are taken on systems. When the randomness of O^n becomes small enough, we know that $x = \arg\min_i V_i^n$ is ruled out in making the nth sampling decision. In the $(n+1)^{th}$ round, since the randomness of O^{n+1} is smaller than that of O^n , if the order of $\{V_i^{n+1}\}$ is the same as the order of $\{V_i^n\}$, $x = \arg\min_i V_i^{n+1} = \arg\min_i V_i^n$ will be ruled out again. Therefore, we will never take sample on $x = \arg\min_i V_i^{n+k} = \arg\min_i V_i^n$ for any $k = 1, 2, ...$ This violates the definition of convergent policy.

Now we give the formal proof of the non-convergence of NKG. We first need a lemma:

Lemma 1. *Let $X = [X_1, X_2]$ be a random vector and $X_1 \neq X_2$ almost surely. If $X^n \to X$ almost surely and $E[|X|] < \infty$. Then $\exists N < \infty$ almost surely such that $\forall n \geq N$, $sgn(X_1^n - X_2^n) = sgn(X_1 - X_2)$ almost surely.*

Proof Define two sets:

$$\mathbb{A}^n := \{\omega \in \Omega : X_2^n(\omega) - X_1^n(\omega) \geq 0, X_1(\omega) - X_2(\omega) \geq 0\}$$
$$\tilde{\mathbb{A}}^n := \{\omega \in \Omega : X_2^n(\omega) - X_2(\omega) + X_1(\omega) - X_1^n(\omega) > 0\}$$

Then $\mathbb{A}^n \subseteq \tilde{\mathbb{A}}^n$ for all $n \geq 1$. Since $X^n \to X$ almost surely, we can have $X_1 - X_1^n + X_2^n - X^2 \to 0$ almost surely. So $P(\bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \tilde{\mathbb{A}}^n) = 0$ and hence $P(\bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \mathbb{A}^n) = 0$.
Define

$$\mathbb{B}^n := \{\omega \in \Omega : X_2^n(\omega) - X_1^n(\omega) < 0, X_1(\omega) - X_2(\omega) \leq 0\}.$$

Similarly, we can show $P(\bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \mathbb{B}^n) = 0$.

Define a random variable:

$$Y^n = \mathbb{I}_{\{X_1^n \leq X_2^n,\ X_1 > X_2\}}$$

since $\{X_1^n \leq X_2^n,\ X_1 \geq X_2\} = \mathbb{A}^n$ we have $Y^n \to 0$ almost surely. Similarly, we can show $Z^n := \{X_1^n > X_2^n,\ X_1 < X_2\} \to 0$ almost surely. Since both Y^n and Z^n are binary random variables, $\exists\ N < \infty$ almost surely such that $Y^n,\ Z^n = 0$ for any $n \geq N$ according to the definition of almost surely convergence. When n is large, we have $Y^n = 0$ and $Z^n = 0$ which implies that $sgn(X_1^n - X_2^n) = sgn(X_1 - X_2)$.

□

Since Frazier et al. (2009) has shown that for any $\pi \in \Pi$ in Lemma A.5., $S^n \to S^\infty$ almost surely, we can easily extend lemma 1 to show that the order of all μ_{ij}^n dost not change after finitely many measurements. This fact is pretty useful in later content.

Now we look at the NKG policy.

Proof of Theorem 2:

For computational clarity, we first assume that all μ_{ij}'s are independent.

We need to show the solution of (19). Let $\{V^n_{(i)}\}$ be the ordered set of $\{V^n_i\}$. Denote equation (19) as $\min_{(x,y)} f(x,y)$. It is obvious that:

$$f(x,y) = \begin{cases} E^n\{V^n_{(1)} \wedge [V^n_x + O^n_x(y)]\}, & \text{if } x \neq \arg\min_i V^n_i \\ E^n\{V^n_{(2)} \wedge [V^n_x + O^n_x(y)]\}, & \text{otherwise} \end{cases} \tag{20}$$

We first assume that $\mu = [\mu_{1:}; \mu_{2:}]$ namely the matrix μ is of two rows. We extend the proof to μ with more rows later. For notational simplicity, we also assume that $V^n_1 \leq V^n_2$. The first case of (20) is less than V^n_1 obviously. We now rewrite the second case of (20) :

$$\begin{aligned} f(1,y) &= E^n\{V^n_2 \wedge [V^n_1 + O^n_1(y)]\} \\ &= \max_{j \neq y} \mu^n_{1j} P(Z < C_1) + E[\mathbb{I}_{\{C_1 \leq Z < C_2\}} \max_j (\mu^n_{1j} + \tilde{\sigma}_j(\Sigma^n_{1,yy}, y)Z)] + V^n_2 P(Z \geq C_2) \end{aligned} \tag{21}$$

where Z is the standard normal random variable and $\{C_i\}$ are the change points of a piecewise linear function.

We now prove the theorem by contradiction. Suppose NKG is a convergent policy since $\mu^n_{ij} \to Y_{ij}$ a.s, then from the previous lemma, there exists an $N < \infty$ almost surely such that the order of each entry of μ^n does not change for all $n > N$. Then exists y^*_1 such that $f(1, y^*_1) > V^n_1$ from direct calculation. From the same calculation, exists y^*_2 such that $f(2, y^*_2) = V^n_1$ because $V^n_2 + O^n_2(y^*_2) > V^n_1$ almost surely. Therefore, $(x^n, y^n) \neq (1, y^*_1)$. Since the order set of $V^1_{\tilde{n}}$ and $V^2_{\tilde{n}}$ remains unchanged for any $\tilde{n} > n$ and $S^{\tilde{n}} \to S^\infty$ according to Lemma A.5 in Frazier et al. (2009), $f(1, y_1; S^{\tilde{n}}) > f(2, y_2; S^{\tilde{n}})$ for any y_1, and y_2 and hence $x^{\tilde{n}} \neq 1$ by induction. This violates the definition of convergent policy and hence leads to contradiction.

Now suppose μ has more than 2 rows. Without loss of generality, we assume that $V^n_1 \leq V^n_i$ for any $i \neq 1$. From the previous analysis, we can derive that $f(1,.) > f(i,.)$ when n is large enough. Therefore, $x^n \neq 1$ for all large n. This means that any system on the first row will be measured a finite number of times even if infinite number of measurements is given.

In the case that μ_{ij}'s are dependent, we suppose that $\mu^n = [\mu^n_{1:}; \mu^n_{2:}]$ and that NKG is convergent. We use tools in large deviation analysis and apply inductioon similar to the independent case to show contradiction.

From the previous lemma, we can further suppose that the order set of μ^n does not change for any n. Without loss of generality, we assume $\max_j \mu^n_{1j} < \max_j \mu^n_{2j}$. Now we define:

$$h(z; \mu, \sigma) = \max_j \{\mu_j + \sigma_j z\}.$$

Let Z be the standard normal distribution. For any μ and σ, $h(z; \mu, \sigma)$ is a piecewise linear map in z. As a result, we know that $h(Z; \mu, \sigma)$ is a sub-Gaussian random variable meaning that the tail of

$h(Z;\mu,\sigma)$ decreases as fast as that of Gaussian distribution. Again, from direct computation, we have:

$$\mathbb{E}[\min_i \max_j \mu^{n+1} \big| S^n, (x^n, y^n) = (2,\cdot)] \le \max_j \mu^n_{1j};$$

$$\begin{aligned}
&\mathbb{E}[\min_i \max_j \mu^{n+1} \big| S^n, (x^n, y^n) = (1,\cdot)] \\
=& \mathbb{E}[\max_j \mu^{n+1}_{1:} \big| S^n, (x^n, y^n) = (1,\cdot)] \\
-& \int_{C^n_1}^{\infty} t d\mathbb{P}\{h(Z;\mu^n_{1:}, \tilde{\sigma}(\mu^n_{1:}, \Sigma^n_{1:,1:})) \le t\} + \max_j \mu^n_{2j} \mathbb{P}\{h(Z;\mu^n_{1:}, \tilde{\sigma}(\mu^n_{1:}, \Sigma^n_{1:,1:})) > C^n_1\} \\
-& \int_{-\infty}^{C^n_2} t d\mathbb{P}\{h(Z;\mu^n_{1:}, \tilde{\sigma}(\mu^n_{1:}, \Sigma^n_{1:,1:})) \le t\} + \max_j \mu^n_{2j} \mathbb{P}\{h(Z;\mu^n_{1:}, \tilde{\sigma}(\mu^n_{1:}, \Sigma^n_{1:,1:})) < C^n_2\}
\end{aligned}$$

where C^n, where $C^n_1 > 0$ and $C^n_2 < 0$, are the intersections of the piecewise linear function $h(z;\mu^n_{1:}, \tilde{\sigma}(\mu^n_{1:}, \Sigma^n_{1:,1:}))$ and constant $\max_j \mu^n_{2j}$. If all the μ_{ij}'s are positively correlated, the last line of the previous equation vanishes. Now since NKG is convergent and from central limit theorem, we know $||\tilde{\sigma}(\mu^n_{1:}, \Sigma^n_{1:,1:})|| \to 0$ in $O(\frac{1}{\sqrt{n}})$ as $n \to \infty$. Therefore, the change points $|C^n| \to \infty$ in $O(\frac{1}{\sqrt{n}})$. Now since $h(z;\mu^n_{1:}, \tilde{\sigma}(\mu^n_{1:}, \Sigma^n_{1:,1:}))$ is sub-Gaussian for any n. With the help of sub-Gaussian concentration theorem, namely $\mathbb{P}\{\big|h(Z;\mu^n_{1:}, \tilde{\sigma}(\mu^n_{1:}, \Sigma^n_{1:,1:}))\big| > t\} \le K_1 e^{-K_2 t^2}$ for some constants K_1 and K_2, we can apply integrate by part to show that

$$\int_{C^n_1}^{\infty} t d\mathbb{P}\{h(Z;\mu^n_{1:}, \tilde{\sigma}(\mu^n_{1:}, \Sigma^n_{1:,1:})) \le t\} \to 0$$

$$\int_{-\infty}^{C^n_2} t d\mathbb{P}\{h(Z;\mu^n_{1:}, \tilde{\sigma}(\mu^n_{1:}, \Sigma^n_{1:,1:})) \le t\} \to 0$$

both in the rate of $O(e^{-n^2})$. On the other hand, $\mathbb{E}[\max_j \mu^{n+1}_{1:} \big| S^n, (x^n, y^n) = (1,\cdot)] - \max_j \mu^n_{1:} \to 0$ in $O(\frac{1}{\sqrt{n}})$. Therefore, exists an N^* such that for any $n \ge N^*$, $\mathbb{E}[\min_i \max_j \mu^{n+1} \big| S^n, (x^n, y^n) = (1,\cdot)] \ge \max_j \mu^n_{1j}$. However, if this happens, only $(2,\cdot)$ will be selected by NKG for all $n \ge N^*$ leading to contradiction.

For μ with more than two rows, we apply the same arguement in the independent case to show that NKG is not convergent. So we can finish the proof.

□

.

One more thing we need to notice is that, by applying the previous sub-Gaussian analysis, we can easily show that the value functions of different sampling decisions converge to the same value in the rate of $O(e^{-n^2})$. In face, when we run simulation of NKG, the algorithm fails to determine the true order set of $\{\mathbb{E}[\min_i \max_j \mu^{n+1} \big| S^n, (x^n, y^n) = (x,y)]\}_{(x,y)}$ numerically after few steps.

A.2. RKG Policy

Benefits of Measurement From equation (2) and (4), we can define a state transition function

$$T(s,(x,y),z):\mathbb{S}\times\{1,\ldots,M\}\times\{1,\ldots,K\}\times\mathbb{R})\mapsto\mathbb{S}$$

such that $T(S^n,(x,y),Z)=S^{n+1}$ where Z is standard normal distributed. We follow the definitions in section 3.2

Before proving optimality properties of RKG, we need the following lemma and propositions. They show that if we provide more measurement opportunities to any stationary measurement policy, then it will perform better on average.

We first prove a lemma which says that the state to which we arrive when (x,y) is measured first and (x',y') second, namely, $T(T(s,(x,y),Z^{n+1}),(x',y'),Z^{n+2})$, equals in distribution to the state to which we measure (x',y') first and (x,y) second, namely $T(T(s,(x',y'),Z^{n+2}),(x,y),Z^{n+1})$. In fact, Frazier et al. (2009) has shown this lemma by logics. Here, we use mathematical proof to make the result more convincing:

Lemma 2. *Given any state* $s=(\mu,\ \Sigma)\in\mathbb{S}$ *and* $(x,y),\ (x',y')\in\{1,2,...,M\}\times\{1,2,...,K\}$, $T(T(s,(x,y),Z^{n+1}),(x',y'),Z^{n+2})$ *equals in distribution to* $T(T(s,(x',y'),Z^{n+2}),(x,y),Z^{n+1})$.

Proof

We first consider the case $x\neq x'$ and without loss of generality, assume $x<x'$. Since $\mu^{n+1}_{x:}$ and $\mu^{n+1}_{x':}$ are independent and $\mu^{n+2}_{x:}$ and $\mu^{n+2}_{x':}$ are independent, we can rewrite the state s as:

$$s=(\mu,\Sigma)=(\mu_{1:},\mu_{2:},...,\mu_{M:},\Sigma_{1:,1:},...,\Sigma_{M:,M:}).$$

According to the definition of T, we have:

$$T(T(s,(x,y),Z^{n+1}),(x',y'),Z^{n+2})=s^{n+2}=(\mu^{n+2},\Sigma^{n+2})$$

with

$$\mu^{n+2}=(\mu_{1:},...,\mu_{x:}+\tilde{\sigma}(\Sigma_{x:,x:},x,y)Z^{n+1},...,\mu_{x':}+\tilde{\sigma}(\Sigma_{x':,x':},x',y')Z^{n+2},...)$$
$$\Sigma^{n+2}=(\Sigma_{1:,1:},...,\Sigma_{x:,x:}-\tilde{\sigma}(\Sigma_{x:,x:},x,y)\tilde{\sigma}(\Sigma_{x:,x:},x,y)^T,...,\Sigma_{x':,x':}-\tilde{\sigma}(\Sigma_{x':,x':},x',y')\tilde{\sigma}(\Sigma_{x':,x':},x',y')^T,...)$$

where

$$\begin{aligned}\tilde{\sigma}(\Sigma_{x:,x:},x,y):&=\frac{\Sigma_{x:,x:}e_y}{\sqrt{\delta^2_{x,y}+\Sigma_{(x,y),(x,y)}}}\\&=\frac{\Sigma_{x:,x:}e_y}{\sqrt{\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y}}\end{aligned}$$

and both Z^{n+1} and Z^{n+2} are standard normal distributed. Obviously, switching the order of (x,y) and (x',y') causes no effect to the distribution of s^{n+2} since they are independent operations on two independent rows and hence on independent elements of s.

When $x=x'$, we first calculate Σ^{n+2}, which is deterministic. Suppose we take sample on (x,y) and then on (x',y'), we have:

$$
\begin{aligned}
\Sigma^{n+2}_{x:,x:} &= \left[\Sigma_{x:,x:} - \tilde{\sigma}(\Sigma_{x:,x:},x,y)\tilde{\sigma}(\Sigma_{x:,x:},x,y)^T\right] \\
&\quad - \tilde{\sigma}(\Sigma_{x:,x:} - \tilde{\sigma}(\Sigma_{x:,x:},x,y)\tilde{\sigma}(\Sigma_{x:,x:},x,y)^T,x',y')\tilde{\sigma}(\Sigma_{x:,x:},x,y)\tilde{\sigma}(\Sigma_{x:,x:},x,y)^T,x',y')^T \\
&= \Sigma_{x:,x:} \\
&\quad - \frac{\Sigma_{x:,x:}e_y e_y^T \Sigma_{x:,x:}}{\delta^2_{x,y} + e_y^T\Sigma_{x:,x:}e_y} - \frac{\Sigma_{x:,x:}e_{y'}e_{y'}^T\Sigma_{x:,x:}}{\delta^2_{x,y'} + e_{y'}^T\Sigma_{x:,x:}e_{y'} - \frac{(e_{y'}^T\Sigma_{x:,x:}e_y)^2}{\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y}} \qquad ① \\
&\quad - \frac{\Sigma_{x:,x:}e_y e_y^T\Sigma_{x:,x:}e_{y'}e_{y'}^T\Sigma_{x:,x:}e_y e_y^T\Sigma_{x:,x:}}{(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)\left[(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)(\delta^2_{x,y'}+e_{y'}^T\Sigma_{x:,x:}e_{y'}) - (e_{y'}^T\Sigma_{x:,x:}e_y)^2\right]} \qquad ② \\
&\quad + 2\frac{\Sigma_{x:,x:}e_y e_y^T\Sigma_{x:,x:}e_{y'}e_{y'}^T\Sigma_{x:,x:}}{(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)(\delta^2_{x,y'}+e_{y'}^T\Sigma_{x:,x:}e_{y'}) - (e_{y'}^T\Sigma_{x:,x:}e_y)^2} \qquad ③
\end{aligned}
$$

We first look at the equation $-①-②$:

$$
\begin{aligned}
-①-② &= \frac{\Sigma_{x:,x:}e_y e_y^T\Sigma_{x:,x:}\left[(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)(\delta^2_{x,y'}+e_{y'}^T\Sigma_{x:,x:}e_{y'}) - (e_{y'}^T\Sigma_{x:,x:}e_y)^2\right]}{(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)\left[(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)(\delta^2_{x,y'}+e_{y'}^T\Sigma_{x:,x:}e_{y'}) - (e_{y'}^T\Sigma_{x:,x:}e_y)^2\right]} \\
&\quad + \frac{(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)^2\Sigma_{x:,x:}e_{y'}e_{y'}^T\Sigma_{x:,x:}}{(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)\left[(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)(\delta^2_{x,y'}+e_{y'}^T\Sigma_{x:,x:}e_{y'}) - (e_{y'}^T\Sigma_{x:,x:}e_y)^2\right]} \\
&\quad + \frac{\Sigma_{x:,x:}e_y e_y^T\Sigma_{x:,x:}e_{y'}e_{y'}^T\Sigma_{x:,x:}e_y e_y^T\Sigma_{x:,x:}}{(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)\left[(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)(\delta^2_{x,y'}+e_{y'}^T\Sigma_{x:,x:}e_{y'}) - (e_{y'}^T\Sigma_{x:,x:}e_y)^2\right]} \\
&= \frac{\delta^2_{x,y}\delta^2_{x,y'}\Sigma_{x:,x:}e_y e_y^T\Sigma_{x:,x:} + \delta^2_{x,y'}\Sigma_{x:,x:}e_y e_y^T\Sigma_{x:,x:}e_y e_y^T\Sigma_{x:,x:}}{(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)\left[(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)(\delta^2_{x,y'}+e_{y'}^T\Sigma_{x:,x:}e_{y'}) - (e_{y'}^T\Sigma_{x:,x:}e_y)^2\right]} \\
&\quad + \frac{\delta^2_{x,y}\Sigma_{x:,x:}e_y e_{y'}^T\Sigma_{x:,x:}e_{y'}e_y^T\Sigma_{x:,x:} + \Sigma_{x:,x:}e_y e_y^T\Sigma_{x:,x:}e_y e_{y'}^T\Sigma_{x:,x:}e_{y'}e_y^T\Sigma_{x:,x:}}{(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)\left[(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)(\delta^2_{x,y'}+e_{y'}^T\Sigma_{x:,x:}e_{y'}) - (e_{y'}^T\Sigma_{x:,x:}e_y)^2\right]} \\
&\quad + \frac{(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)(\delta^2_{x,y}\Sigma_{x:,x:}e_{y'}e_{y'}^T\Sigma_{x:,x:} + \Sigma_{x:,x:}e_{y'}e_y^T\Sigma_{x:,x:}e_y e_{y'}^T\Sigma_{x:,x:})}{(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)\left[(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)(\delta^2_{x,y'}+e_{y'}^T\Sigma_{x:,x:}e_{y'}) - (e_{y'}^T\Sigma_{x:,x:}e_y)^2\right]} \\
&= \frac{\delta^2_{x,y}\Sigma_{x:,x:}e_{y'}e_{y'}^T\Sigma_{x:,x:} + \Sigma_{x:,x:}e_{y'}e_y^T\Sigma_{x:,x:}e_y e_{y'}^T\Sigma_{x:,x:}}{(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)(\delta^2_{x,y'}+e_{y'}^T\Sigma_{x:,x:}e_{y'}) - (e_{y'}^T\Sigma_{x:,x:}e_y)^2} \\
&\quad + \frac{\delta^2_{x,y'}\Sigma_{x:,x:}e_y e_y^T\Sigma_{x:,x:} + \Sigma_{x:,x:}e_y e_{y'}^T\Sigma_{x:,x:}e_{y'}e_y^T\Sigma_{x:,x:}}{(\delta^2_{x,y}+e_y^T\Sigma_{x:,x:}e_y)(\delta^2_{x,y'}+e_{y'}^T\Sigma_{x:,x:}e_{y'}) - (e_{y'}^T\Sigma_{x:,x:}e_y)^2}
\end{aligned}
$$

We can see that replacing all y by y' and all y' by y causes no effect to $①+②$. That is, the equation is symmetric in y and y'. This property also holds for ③. So the matrix $\Sigma^{n+2}_{x:,x:}$ is symmetric in y and y'. For any $i \neq x$, $\Sigma^{n+2}_{i:,i:} = \Sigma_{i:,i:}$, so we know switching the order of sampling causes no effect to Σ^{n+2}.

Now we show the transition function from μ^n to μ^{n+2} if we take sample on (x,y) first and then on (x',y'):

$$\mu^{n+2}_{x:} = \mu_{x:} + \tilde{\sigma}(\Sigma_{x:,x:},x,y)Z^{n+1} + \tilde{\sigma}(\Sigma_{x:,x:} - \tilde{\sigma}(\Sigma_{x:,x:},x,y)\tilde{\sigma}(\Sigma_{x:,x:},x,y)^T,x',y')Z^{n+2}$$

where Z^{n+1} and Z^{n+2} are standard normal distributed. Then we define a random variable:

$$Y := \tilde{\sigma}(\Sigma_{x:,x:},x,y)Z^{n+1} + \tilde{\sigma}(\Sigma_{x:,x:} - \tilde{\sigma}(\Sigma_{x:,x:},x,y)\tilde{\sigma}(\Sigma_{x:,x:},x,y)^T,x',y')Z^{n+2}.$$

Since Y is the sum of two normal distributed random variable, it is also normal distributed with mean 0. The covariance matrix of Y is of the form

$$\begin{aligned}
&\tilde{\sigma}(\Sigma_{x:,x:},x,y)\tilde{\sigma}(\Sigma_{x:,x:},x,y)^T \\
&+\tilde{\sigma}(\Sigma_{x:,x:} - \tilde{\sigma}(\Sigma_{x:,x:},x,y)\tilde{\sigma}(\Sigma_{x:,x:},x,y)^T,x',y')\tilde{\sigma}(\Sigma_{x:,x:} - \tilde{\sigma}(\Sigma_{x:,x:},x,y)\tilde{\sigma}(\Sigma_{x:,x:},x,y)^T,x',y')^T \\
&= -(\textcircled{1}+\textcircled{2}+\textcircled{2})
\end{aligned}$$

This equality is from previous calculations. So we know that replacing all y by y' and all y' by y causes no effect to the covariance matrix of Y. Moreover, for any $i \neq x$, $\mu^{n+2}_{i:} = \mu_{i:}$. Therefore, switching the order of sampling causes no effect to the transition function from s^n to s^{n+2}

□

Now we state the following proposition which says that measurement always give improvement in expectation:

Proposition 3. *$Q^n(s,x,y) \leq V^{n+1}(s)$ for every $0 \leq n < N$, $s \in \mathbb{S}$, and $(x,y) \in \{1,...,M\} \times \{1,...K\}$*

Proof

We prove by backward induction on n. When n=N-1, for any $s \in \mathbb{S}$ we have

$$\begin{aligned}
Q^{N-1}(s,x,y) &= \mathbb{E}[\min_i \mathbb{E}[\max_j \theta_{ij} \Big| \mathcal{F}^N] \Big| S^{N-1} = s, (x^N,y^N) = (x,y)] \\
&\leq \min_i \mathbb{E}[\mathbb{E}[\max_j \theta_{ij} \Big| \mathcal{F}^N] \Big| S^{N-1} = s, (x^N,y^N) = (x,y)] \\
&= c \wedge \mathbb{E}[\max\{(\Sigma^{N-1}_{x:,x:} - \tilde{\sigma}(\Sigma^{N-1},x,y)\tilde{\sigma}(\Sigma^{N-1},x,y)^T)^{\frac{1}{2}} Z_K + \mu^{N-1}_{x:} + \tilde{\sigma}(\Sigma^{N-1},x,y)Z_1\}] \\
&= c \wedge \mathbb{E}[\max\{(\Sigma^{N-1}_{x:,x:})^{\frac{1}{2}} Z_K + \mu^{N-1}_{x:}\}] \\
&= \min_i \mathbb{E}[\max_j \theta_{ij} \Big| S^{N-1} = s] = V^N(s).
\end{aligned}$$

where

$$\begin{aligned}
&c = \min_{i \neq x} \mathbb{E}[\max_j \theta_{ij} \Big| S^{N-1} = s] \\
&Z_k \sim \mathcal{N}(0, \mathbb{I}_k) \quad \text{(standard normal in } \mathbb{R}^k)
\end{aligned}$$

The first inequality is due to Jensen's inequality and the concavity of min function;the fourth line is from direct calculation. Now suppose the inequality $Q^{n+1}(s,x,y) \le V^{n+2}(s)$ holds for every $s \in \mathbb{S}$, and $(x,y) \in \{1,...,M\} \times \{1,...K\}$ then we have:

$$\begin{aligned}
Q^n(s,x,y) &= \mathbb{E}[V^{n+1}(T(s,(x,y),Z^{n+1}))] \\
&= \mathbb{E}[\min_{(x',y')} Q^{n+1}(T(s,(x,y),Z^{n+1}),x',y')] \\
&\le \min_{(x',y')} \mathbb{E}[Q^{n+1}(T(s,(x,y),Z^{n+1}),x',y')] \\
&= \min_{(x',y')} \mathbb{E}[V^{n+2}(T(T(s,(x,y),Z^{n+1}),(x',y'),Z^{n+2}))]
\end{aligned}$$

According to the previous lemma, we know that the state to which we arrive when (x,y) is measured first and (x',y') second, namely, $T(T(s,(x,y),Z^{n+1}),(x',y'),Z^{n+2})$, equals in distribution to the state to which we measure (x',y') first and (x,y) second, namely $T(T(s,(x',y'),Z^{n+2}),(x,y),Z^{n+1})$.

Therefore, we have the following equations

$$\begin{aligned}
Q^n(s,x,y) &\le \min_{(x',y')} \mathbb{E}[V^{n+2}(T(T(s,(x,y),Z^{n+1}),x',y',Z^{n+2}))] \\
&= \min_{(x',y')} \mathbb{E}[V^{n+2}(T(T(s,(x',y'),Z^{n+2}),x,y,Z^{n+1}))] \\
&= \min_{(x',y')} \mathbb{E}[\mathbb{E}[V^{n+2}(T(T(s,(x',y'),Z^{n+2}),(x,y),Z^{n+1}))|Z^{n+2}]] \\
&= \min_{(x',y')} \mathbb{E}[Q^{n+1}(T(s,(x',y'),Z^{n+2}),x,y)] \\
&\le \min_{(x',y')} \mathbb{E}[V^{n+2}(T(s,(x',y'),Z^{n+2}))] \\
&= \min_{(x',y')} Q^{n+1}(s,x',y') \\
&= V^{n+1}(s),
\end{aligned}$$

where the fifth line is from the induction hypothesis. So we have $Q^n(s,x) \le V^{n+1}(s)$ and thus the proof is finished.

□

A policy π is said to be *stationary* if it is independent of time n. The following proposition shows that for any stationary policy, the value function decreases as more measurement number is allowed.

PROPOSITION 4. *For any stationary policy π and state $s \in \mathbb{S}$, $V^{n,\pi}(s) \le V^{n+1,\pi}(s)$*

Proof

We prove by backward induction on n.For the base case $n = N-1$:

$$\begin{aligned}
V^{N-1,\pi}(s) &= \mathbb{E}^{\pi}[V^N(S^N)|S^{N-1}=s] \\
&= \mathbb{E}^{\pi}[\min_i \mathbb{E}[\max_j \theta_{ij}|\mathcal{F}^N]|S^{N-1}=s] \\
&\le \min_i \mathbb{E}^{\pi}[\mathbb{E}[\max_j \theta_{ij}|\mathcal{F}^N]|S^{N-1}=s] \\
&= \min_i \mathbb{E}[\max_j \theta_{ij}|s] = V^N(s)
\end{aligned}$$

The third line is justified by Jensen's inequality and the last line is because $V^{N,\pi}(s)$ is independent of π. Suppose the inequality $V^{n+1,\pi}(s) \leq V^{n+2,\pi}(s)$ holds for every $s \in \mathbb{S}$, then according to definition we have

$$\begin{aligned} V^{n,\pi}(s) &= \mathbb{E}[V^{n+1,\pi}(T(s, \mathcal{A}^{\pi}(s), Z^{n+1}))] \\ &\leq \mathbb{E}[V^{n+2,\pi}(T(s, \mathcal{A}^{\pi}(s), Z^{n+1}))] \\ &= V^{n+1,\pi}(s) \end{aligned}$$

where the last line is by definition. The proof is finished.

□

Corollary 1. *For every $s \in \mathbb{S}$, $V^n(s) \leq V^{n+1}(s)$.*

Proof

In proposition 1, since the inequality holds for every (x, y), we have
$V^n(s) = \min_{(x,y)} Q^n(s, x, y) \leq V^{n+1}(s)$.

□

Convergence and asymptotic optimality With Benefits of measurement now we can prove the convergence property and asymptotic optimality. On its own, convergence or asymptotic optimality of a policy is irrelevant to convergence rate in the finite sampling budget case. EA or MV are convergent and asymptotically optimal according to proposition 1 but they do not have acceptable performance as shown by numerical experiments. Convergence and asymptotic optimality of a policy ensure that it can ultimately gives the correct result if enough amount of sampling budget is given. They are the required conditions of robustness.

We first define the asymptotic optimal value by $V(s; \infty) := \lim_{N\to\infty} V^0(s; N)$ and the asymptotic value of a policy π by $V^{\pi}(s; \infty) := \lim_{N\to\infty} V^{0,\pi}(s; N)$. We first show the existence and boundedness of $V(s; \infty)$.

Proposition 5. *$\forall s \in \mathbb{S}$, $V(s; \infty)$ and for every stationary policy π, $V^{\pi}(s; \infty)$ exist and is bounded below by*

$$U(s) := \mathbb{E}[\min_i \max_j \theta_i j | S^0 = s] > -\infty$$

Further, $V^{\pi}(s; \infty)$ is finite and bounded below by $U(s)$ for any policy π.

Proof

We can, from Markov property of every measurement, derive that for every initial state $s^0 \in \mathbb{S}$, $V^0(s^0; N-1) = V^1(s^0; N)$. So, by induction and corollary 1 we have $\forall s^0 \in \mathbb{S}$, $V^0(s^0, N)$ is a non-decreasing function in N.

We can also show that $V^{\pi,0}(s^0;N)$ is a non-decreasing function in N in a similar way by applying proposition 4.

Now we show that $\forall N \geq 1$, $\forall s^0 \in \mathbb{S}$, $V^0(s^0;N) \geq U(s^0)$.

For every $\pi \in \Pi$, we have

$$\begin{aligned}\mathbb{E}^\pi[\min_i \mathbb{E}[\max_j \theta_{ij}|\mathcal{F}^N]|S^0 = s^0] &\geq \mathbb{E}^\pi[\mathbb{E}[\min_i \max_j \theta_{ij}|\mathcal{F}^N]|S^0 = s^0] \\ &= \mathbb{E}[\min_i \max_j \theta_{ij}|S^0 = s^0] = U(s^0)\end{aligned}$$

So by letting $N \to \infty$, we have for every policy $\pi \in \Pi$

$$\forall s^0 \in \mathbb{S}, \quad V^\pi(s^0;\infty) \geq V(s^0;\infty) \geq U(s^0)$$

Since both $V^\pi(s;N)$ and $V^0(s;N)$ are monotone and bounded from below for fixed s, $V(s;\infty)$ and $V^\pi(s;\infty)$ exist and are bounded.

□

We now introduce some lemmas which are useful for the proof of main result.

Lemma 3. *The sequence of states $\{S^n\}$ converges almost surely to a random variable S^∞ in $\mathbb{S}$*

This Lemma can be easily proved by generalizing Lemma A.6. in Frazier et al. (2009) and using Lemma 5.5 and Theorem 3.12 in Kallenberg (1997). We skip the proof here.

Lemma 4. *Let (Ω, Σ, μ) be a probability space, $\{X^n\}$ and X be (Ω, Σ, μ)-measurable functions, and $f : \mathbb{R}^m \mapsto \mathbb{R}$ be a continuous Lipschitz function with $f(0) = 0$. If $X^n \to X$ in L^p where $1 \leq p < \infty$, then $f(X^n) \to f(X)$ in L^p.*

Lemma 4 directly follows from Theorem III.3.6 and III.9.1 in Dunford and Schwartz (2009) or from Theorem 6 in Bartle and Joichi (1961).

We now can prove proposition 1 with ease.

Proof of Proposition 1:

We have assumed in the formal model in section 3 that $\{\max_j \theta_{ij}\}_{i=1}^K$ are integrable. By applying Theorem (5.6) in section 4.5, Durrett (2005), we can immediately show that, $\forall i \in \{1, \ldots, K\}$, $\mathbb{E}[\max_j \theta_{ij}|\mathcal{F}^N] \to \max_j \theta_{ij}$ almost surely and in L^1 given that the filtration $\{\mathcal{F}^n\}_{n=0}^\infty$ is generated by a convergent policy.

Now we prove that $\max_i\{\cdot\}$ and $\min_i\{\cdot\}$ are Lipschitz functions. We just need to prove $\max_i\{\cdot\}$ is lipschitz. The proof of $\min_i\{\cdot\}$ is similar. Firstly, we have the following inequality:

$$\left|\max_i\{x_i\} - \max_i\{y_i\}\right| \leq \max_i\{|x_i - y_i|\} = ||x - y||_\infty.$$

We then apply Holder inequality to get

$$||x-y||_\infty \le ||x-y||.$$

Therefore, $\max_i\{\cdot\} \le ||x-y||$ for any x and y. Moreover, $\max_i\{0\}=0$. This implies, via Lemma 4, that $\min_i \mathbb{E}[\max_j \theta_{ij}|\mathcal{F}^N] \to \min_i \max_j \theta_{ij}$ in L^1. Convergence in L^1 implies

$$V^\pi(S^0;\infty) = \lim_{N\to\infty} \mathbb{E}^\pi[\min_i \mathbb{E}[\max_j \theta_{ij}|\mathcal{F}^N]] = \mathbb{E}[\min_i \max_j \theta_{ij}] = U(S^0)$$

□

Before the proof of proposition 2, we give one more lemma, which is a more general version of Lemma A.7 in Frazier et al. (2009).

Lemma 5. *$Q^{N-1}(S^\infty,x,y) = V^N(S^\infty)$ almost surely under π for any (x,y) if and only if the policy π samples each system (x,y) infinitely often.*

Proof : The if direction is almost the same as lemma A.7. We only need to modify the Q-factor and value function then we are done. Now we prove the only if direction. Suppose $Q^{N-1}(S^\infty,x,y) = V^N(S^\infty)$ amost surely for any (x,y). The following equality holds almost surely:

$$\begin{aligned}&\min_i \mathbb{E}[\max_j \theta_{ij}\Big|S^\infty]\\ =&\mathbb{E}[\min_i \mathbb{E}[\max_j \theta_{ij}\Big|\mu^\infty_{x:} + \tilde{\sigma}(\Sigma^\infty,x,y)Z,\ \Sigma^\infty_{x:,x:} - \tilde{\sigma}(\Sigma^\infty,x,y)\tilde{\sigma}(\Sigma^\infty,x,y)^T]\Big|S^\infty]\end{aligned}$$

where $Z\sim\mathcal{N}(0,1)$. If $\tilde{\sigma}(\Sigma^\infty,x,y)\neq 0$, then there must exists $\omega\in\Omega$ such that $S^\infty(\omega)=S=(\mu,\Sigma)$

$$\begin{aligned}&\min_i \mathbb{E}[\max_j \theta_{ij}\Big|S]\\ \neq&\mathbb{E}[\min_i \mathbb{E}[\max_j \theta_{ij}\Big|\mu_{x:} + \tilde{\sigma}(\Sigma,x,y)Z,\ \Sigma_{x:,x:} - \tilde{\sigma}(\Sigma,x,y)\tilde{\sigma}(\Sigma,x,y)^T]\Big|S].\end{aligned}$$

Then since $V^N(s)$ is continuous in s, we can derive that $\exists B_\epsilon(S)\subset\mathbb{S}$ with $\epsilon>0$, such that $\forall s\in B_\epsilon(S)$, the previous inequality holds. However, this implies that $\mathbb{P}\{Q^{N-1}(S^\infty,x,y)=V^N(S^\infty)\}<1$ which is impossible. Therefore, $\tilde{\sigma}(\Sigma^\infty,x,y)=0$ for any (x,y). So we have $||\Sigma^\infty||=0$. According to theorem 1, $||\Sigma^\infty||=0$ if and only if the policy π samples each system (x,y) infinitely often.

□

The proof of RKG's convergence property is almost the same as Theorem 4 in Frazier et al. (2009) since we have established a similar framework for proving the convergence result. We briefly sketch the proof first. We prove proposition 2 by contradiction. If RKG is not convergent then by proposition 1 and lemma 3 there exists a set A such that $\forall(x,y)\in A$, $Q^{N-1}(S^\infty,x,y)<V^N(S^\infty)$ and $\forall(x,y)\notin A$, $Q^{N-1}(S^\infty,x,y)=V^N(S^\infty)$. This leads to contradiction due to the nature of RKG.

Proof of Proposition 2:

From Lemma 3, The sequence of states $\{S^n\}$ generated by RKG converge to a random variable S^∞ almost surely. We first denote (i,j) by a vector $x=(x_1,x_2)$ for simplicity.

Consider the event $H_x := \{Q^{N-1}(S^\infty,x) < V^N(S^\infty)\}$ where $x \in \{1,...,K\} \times \{1,...,M\}$. Let $A \subset \{1,...,K\} \times \{1,...,M\}$. Suppose A consists of all the positions on which RKG only takes finite samples. We define

$$H_A := \{\cap_{x\in A} H_x\} \cap \{\cap_{x\notin A} H_x^C\}$$

where H_x^C is the complement of H_x. From proposition 3 we know $Q^{N-1}(s;x) \le V^N(s)$ for every $s \in \mathbb{S}$ so, according to lemma 5, $\forall \omega \in H_A$, $Q^{N-1}(S^\infty(\omega);x) < V^N(S^\infty(\omega))$ for every $x \in A$ and $Q^{N-1}(S^\infty(\omega);x) = V^N(S^\infty(\omega))$ for every $x \notin A$. We now show that for any such A, if $A \neq \emptyset$, $P(H_A)=0$.

Let $K_x(\omega) < \infty$ where $x \in A$ and $\omega \in \Omega := H_A \cap \{\omega : S^n(\omega) \to S^\infty(\omega\}$ be the number of times that RKG takes sample on position x. Let $K(\omega) = \max_x K_x(\omega)$. So RKG never samples positions in A after K:

$$x^n(\omega) \notin A \quad \forall \omega \in \Omega \ , n > K(\omega).$$

However, if such an A is not empty, we can derive that $\forall x \in A$, $Q^{N-1}(S^\infty(\omega);x) < V^N(S^\infty(\omega)) = \min_{y\in A^C} Q^{N-1}(S^\infty(\omega);y)$. So there exists another $\tilde{n} > K(\omega)$ with probability 1 that $Q^{N-1}(S^{\tilde{n}}(\omega);x) < \max_{y\in A^C} Q^{N-1}(S^{\tilde{n}}(\omega);y)$. RKG is going to choose $x \in A$ in this situation which contradicts the assumption. As a result we have $P(H_A)=0$ which means there is no x that RKG samples finitely often.

□

Theorem 3 can be easily derived from proposition 1 and proposition 2.

Suboptimal Bound We have shown that RKG is optimal for $N=1$ and $N=\infty$. We now prove Theorem 4 which gives the suboptimal bound of the policy. We first present the following lemma, which give some useful estimate if future calculation.

LEMMA 6. *If $\theta \sim \mathcal{N}(\mu,\Sigma)$ is a multivariate Gaussian random variable on $\mathbb{R}^m$ then* $\mathbb{E}[\max_i\{\theta_i\}] - \max_i\{\mu_i\} \le \sqrt{2||\Sigma||\log m}$

For any s,

$$\begin{aligned}\exp\{s\left(\mathbb{E}[\max_i \theta_i] - \max_j \mu_j\right)]\} &\le \mathbb{E}[\exp\{s(\max_i \theta_i - \max_j \mu_j)\}] \\ &= \mathbb{E}[\max_i\{\exp\{s(\theta_i - \max_j \mu_j)\}\}]\end{aligned}$$

$$
\begin{aligned}
&\leq \sum_{i=1}^{m} \mathbb{E}[\exp\{s(\theta_i - \max_j \mu_j)\}] \\
&\leq \sum_{i=1}^{m} \mathbb{E}[\exp\{s(\theta_i - \mu_i)\}] \\
&= \sum_{i=1}^{m} \exp\{\frac{1}{2}\sigma_{ii}^2 s^2\} \\
&\leq m \exp\{\frac{1}{2}||\Sigma|| s^2\}
\end{aligned}
$$

where the first line holds because of Jensen's inequality. We then take *log* on both sides and let $s = \frac{\sqrt{2\log m}}{\sqrt{||\Sigma||}}$ to get the result.

□

The core of Theorem 4 is contained in the following lemma, which bounds the marginal value of the last sampling $x^{N-1} = (x_1^{N-1}, x_2^{N-1})$.

LEMMA 7. *Let* $s = (\mu, \Sigma) \in \mathbb{S}$, *Then*

$$
V^{N-1}(s) \geq V^N(s) - [\frac{1}{\sqrt{2\pi}}||\tilde{\sigma}(\Sigma, \cdot)|| + \sqrt{2||\Sigma_{\cdot:,\cdot:}||\log K}] \tag{22}
$$

Proof

Bellman's equation implies $V^{N-1}(s) = \min_{x^{N-1}} \mathbb{E}[V^N(S^N)|S^{N-1} = s]$. We can bound $V^N(S^N)$ by:

$$
\begin{aligned}
V^N(S^N) &= \min_i \mathbb{E}[\max_j \theta_{ij}|S^N] \\
&= c \wedge \mathbb{E}[\max_j \theta_{x_1^{N-1}j}|S^N] \\
&\geq c \wedge \max_j \{\mu^{N-1}_{x_1^{N-1}j} + \tilde{\sigma}_j(\Sigma^{N-1}, x^{N-1})Z^N\} \\
&\geq c \wedge \max_j \{\mu^{N-1}_{x_1^{N-1}j}\} - \left|\max_j \tilde{\sigma}_j(\Sigma^{N-1}, x^{N-1})Z^N\right| \\
&\geq c \wedge \{\mathbb{E}[\max_j \{\theta_{x_1^{N-1}j}\}] - \sqrt{2||\Sigma^{N-1}_{\cdot:,\cdot:}||\log K}\} - \left|\max_j \tilde{\sigma}_j(\Sigma^{N-1}, x^{N-1})Z^N\right| \\
&\geq V^N(S^{N-1}) - \sqrt{2||\Sigma^{N-1}_{\cdot:,\cdot:}||\log K} - \left|\max_j \tilde{\sigma}_j(\Sigma^{N-1}, x^{N-1})Z^N\right|
\end{aligned} \tag{23}
$$

where

$$
c = \min_{i \neq x_1^{N-1}} \mathbb{E}[\max_j \theta_{ij}|S^{N-1}].
$$

The third line of equation (23) is from Jensen's inequality; the fifth line is from Lemma 6; the last line is because $\sqrt{2||\Sigma^{N-1}_{\cdot:,\cdot:}||\log K}$ is non-negative.

So we can bound $V^{N-1}(s)$ by

$$
\begin{aligned}
V^{N-1}(s) &\geq \min_{x^{N-1}} \mathbb{E}[V^N(S^{N-1}) - \sqrt{2||\Sigma_{\cdot:,\cdot:}||\log K} - \left|\max_j \tilde{\sigma}_j(\Sigma, x^{N-1})Z^N\right|] \\
&\geq V^N(s) - \max_{x^{N-1}} \{\sqrt{2||\Sigma_{x_1^{N-1}:,x_1^{N-1}:}||\log K} + \mathbb{E}[\left|\max_j \tilde{\sigma}_j(\Sigma, x^{N-1})Z^N\right| \Big| S^{N-1} = s]\}
\end{aligned}
$$

After some steps of calculation, we have

$$\mathbb{E}[\big|\max_j \tilde{\sigma}_j(\Sigma, x^{N-1})Z^N\big|\big|S^{N-1}=s] = [\max_j \tilde{\sigma}_j(\Sigma, x^{N-1}) + \min_j \tilde{\sigma}_j(\Sigma, x^{N-1})]\mathbb{E}[Z^+] = \frac{1}{\sqrt{2\pi}}||\tilde{\sigma}(\Sigma,\cdot)||$$

. We substitute this identity into the previous inequality, completing the proof.

□

We extend the bound shown in Lemma 7 to hold when there are more number of sampling opportunities remaining.

Proposition 6.

$$V^n(S^n) \geq V^{N-1}(S^n) - \max_{x^n,\ldots,x^{N-2}}\{\sum_{k=n+1}^{N-1}\frac{1}{\sqrt{2\pi}}||\tilde{\sigma}(\Sigma^k,\cdot)|| + \sqrt{2||\Sigma^k_{\cdot:,\cdot:}||\log K}\}$$

Proof

When in the state s^n, Σ^{n+1} is a deterministic function of x^n and S^n and, by induction, Σ^{N-1} is a deterministic function of $\{x^n,...,x^{N-2}\}$ and S^n. Now we proof the proposition by backward induction. The base case $n = N-1$ is obviously true. By Bellman's equation and the induction hypothesis, we have

$$\begin{aligned} V^n(S^n) &= \min_{x^n} E[V^{n+1}(S^{n+1})|S^n] \\ &\geq \min_{x^n} E[V^{N-1}(S^{n+1}) - \max_{\{x^{n+1},\ldots,x^{N-2}\}}\{\sum_{k=n+2}^{N-1}\frac{1}{\sqrt{2\pi}}||\tilde{\sigma}(\Sigma^k,\cdot)|| + \sqrt{2||\Sigma^k_{\cdot:,\cdot:}||\log K}\}\big|S^n] \end{aligned}$$

Now we prove that the inequality holds for n. We apply Lemma 7 to $V^{N-1}(S^{n+1})$, we have:

$$\begin{aligned} V^n(S^n) &\geq \min_{\tilde{x}^n} E[V^N(S^{n+1}) - \max_{\{x^n,\ldots,x^{N-2}\}}\{\sum_{k=n+2}^{N-1}\frac{1}{\sqrt{2\pi}}||\tilde{\sigma}(\Sigma^k,\cdot)|| + \sqrt{2||\Sigma^k_{\cdot:,\cdot:}||\log K}\}\big|S^n] \\ &\geq \min_{x^n} E[V^N(S^{n+1})\big|S^n] - \max_{\{x^n,\ldots,x^{N-2}\}}\{\sum_{k=n+2}^{N-1}\frac{1}{\sqrt{2\pi}}||\tilde{\sigma}(\Sigma^k,\cdot)|| + \sqrt{2||\Sigma^k_{\cdot:,\cdot:}||\log K}\} \end{aligned}$$

Noting that the first term on the right-hand side is, in fact, $V^{N-1}(S^n)$ shows the result.

□

Finally, we can prove Theorem 4 by applying Lemma 7 and Proposition 6.

Proof of Theorem 4:

The RKG policy is optimal when $N = 1$ by definition, we have $V^{N-1}(S^n) = V^{N-1,\pi^{RKG}}(S^n)$. From benefits of measurement, we have $V^{n,\pi^{RKG}}(S^n) \leq V^{N-1,\pi^{RKG}}(S^n)$. Substituting these inequalities into Proposition 6 shows the result.

□

MC Estimate of RKG In general, MC estimation lowers the performance of a policy. However, we can show that all the optimality results remain if its MC estimation gives non-zero correct probability. We only need to prove Theorem 5 for Theorem 6 holds true because of the nature of convergent policy.

Proof of Theorem 5:

We assume that $\hat{\pi}$ is not convergent and show that the assumption leads to contradiction. Let $A \subset \{1,\dots,M\} \times \{1,\dots,K\}$ be the set of positions on which $\hat{\pi}$ samples finitely often. By lemma 3, we have $S^n(\hat{\pi}) \to S^\infty(\hat{\pi})$ almost surely.

Define

$$H_x = \{\omega : \sum_{n=1}^{\infty} \mathbb{I}_{\{\mathcal{A}^\pi(S^n(\hat{\pi}))=x\}} < \infty\}$$

where $\mathcal{A}^\pi$ is the decision function of π. Further, denote $H_A := \cup_{x\in A} H_x$. H_A is the set of events that π, the true policy, does not try to make correction on wrong decision by $\hat{\pi}$. Because $\hat{\pi}$ samples any alternative in A finitely often and $\{S^n(\hat{\pi})\}_{n=0}^\infty$ is the sequence of states induced by $\hat{\pi}$, there must be a large number N^* such that $\mathcal{A}^{\hat{\pi}}(S^n(\hat{\pi})) \notin A$ for any $n \geq N^*$. Now suppose we can compute the decision function of the true policy, namely $\mathcal{A}^\pi$. If we plug all the states $\{S^n(\hat{\pi})\}_{n=0}^\infty$ in $\mathcal{A}^\pi$, then as $\hat{\pi}$ stops sampling on A and keeps sampling on A^C, benefit of measurement on A^C vanishes. Therefore, the true policy π will try to make correction $\mathcal{A}^\pi(S^n(\hat{\pi})) \in A$ for some large n. H_A is the set of events that number of corrections made is finite. We will show $\mathbb{P}(H_A) = 0$ if A is not empty.

We mimic the proof of Proposition 2 to show that $\mathbb{P}(H_A) = 0$. Let $\omega \in H_A \cap \{S^n(\hat{\pi}) \to S^\infty(\hat{\pi})\}$. Suppose $\mathbb{P}(H_A) > 0$ and A is not empty, then for any $x \in A$ and any $y \notin A$, we have $Q^{N-1}(S^\infty(\omega), x) \geq V^N(S^\infty(\omega)) > Q^{N-1}(S^\infty(\omega), y)$ which simply means that alternative y is with benefits of measurement and x is not. However, since $\omega \in \{S^n(\hat{\pi}) \to S^\infty(\hat{\pi})\}$ and y has been sampled finitely often at state $S^\infty(\hat{\pi})$, we know that $Q^{N-1}(S^\infty(\omega), y) = V^N(S^\infty(\omega))$. On the other hand, since x has been sampled a finite number of times at $S^\infty(\omega)$, we can derive that $Q^{N-1}(S^\infty(\omega), x) \leq V^N(S^\infty(\omega))$. This leads to contradiction and hence $\mathbb{P}(H_A) = 0$ if A is not empty. Therefore, as S^n converges to S^∞, RKG has made infinite number of sampling decisions on $x \in A$ but its MC estimator fails to select x infinitely often after finite steps.

More precisely, we write down the mathematical expression of our result. Denote

$$n_k = \inf\{n : \sum_{n=1}^{n} \mathbb{I}_{\{\mathcal{A}^\pi(S^n(\hat{\pi}))\in A\}} = k\}.$$

n_k is the number that policy π selects alternative in A k times if we plug $S^1(\hat{\pi}), ..., S^{n_k}(\hat{\pi})$ in $\mathcal{A}^\pi$ one by one. Because $\mathbb{P}(H_A^C) = 1$, we know that $\forall k > 0$, n_k exists and $\mathbb{P}(n_k < \infty) = 1$.

When we apply consistent Monte Carlo estimation, for any $s \in \mathbb{S}$, we have:

$$\mathbb{P}\{\mathcal{A}^{\hat{\pi}}(s) = \mathcal{A}^\pi(s)\} = \prod_{(x,y)\neq(x^*,y^*)} \mathbb{P}\{\frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x^*, y^*; s) < \frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x, y; s)\}$$

where L is the number of MC samples drawn and

$Y^{(k)}(x,y;s)$ is the kth MC sample drawn to estimate $\mathbb{E}\big[\min_i \mathbb{E}[\max_j \theta_{ij}|S^{n+1}]\big|S^n=s,(x^n,y^n)=(x,y)\big]$

$(x^*,y^*)=\arg\min_{(x,y)} \mathbb{E}\left[\min_i \mathbb{E}[\max_j \theta_{ij}|S^{n+1}]\big|S^n=s,(x^n,y^n)=(x,y)\right]$.

Since the MC estimator is consistent, for any $s\in\bar{\mathbb{S}}$, the closure of $\mathbb{S}$, each term in the previous equation satisfies:

$$\lim_{L\to\infty} \mathbb{P}\{\frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x^*,y^*;s) < \frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x,y;s)\}=1. \tag{24}$$

So if there exists some L such that

$$\inf_{s\in\mathbb{S}} \mathbb{P}\{\frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x^*,y^*;s) < \frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x,y;s)\}=0$$

then L must be finite. For the case L=1, this is impossible because, under our framework, both $Y^{(k)}(x^*,y^*;s)$ and $Y^{(k)}(x,y;s)$ must be continuously and positively distributed on the whole domain almost everywhere for any $s=(\mu,\Sigma)\in\mathbb{S}$. Then for $s\in\mathbb{S}$, we can prove by induction that for finite L, the convolution of continuous distributions is still continuously and positively distributed on the whole domain. For $s\in\bar{\mathbb{S}}-\mathbb{S}$, we only need to check two cases, either $||\mu||\to\infty$ or $||\Sigma||\to\infty$ but none of them satisfies the previous equality.

Therefore, for any $L<\infty$, we have :

$$\inf_{s\in\mathbb{S}} \mathbb{P}\{\frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x^*,y^*;s) < \frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x,y;s)\}>0.$$

As a result, by putting two cases together, we have:

$$\inf_{s\in\mathbb{S}} \mathbb{P}\{[\frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x^*,y^*;s) < \frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x,y;s)]\}>0$$

and hence:

$$\inf_{s\in\mathbb{S}} \mathbb{P}\{\mathcal{A}^{\hat{\pi}}(s)=\mathcal{A}^{\pi}(s)\} \geq \prod_{(x,y)\neq(x^*,y^*)} \inf_{s\in\mathbb{S}} \mathbb{P}\{\frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x^*,y^*;s) < \frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x,y;s)\}>0$$

and hence

$$\sup_{s\in\mathbb{S}} \mathbb{P}\{\mathcal{A}^{\hat{\pi}}(s)\neq\mathcal{A}^{\pi}(s)\}<1.$$

Now since we have made sure that the probability of making wrong estimation must be strictly less than 1, we can derive from dominated convergence theorem that

$$\lim_{n\to\infty} \mathbb{P}\{\mathcal{A}^{\hat{\pi}}(S^n(\hat{\pi}))\neq\mathcal{A}^{\pi}(S^n(\hat{\pi}))\}=\mathbb{P}\{\mathcal{A}^{\hat{\pi}}(S^\infty(\hat{\pi}))\neq\mathcal{A}^{\pi}(S^\infty(\hat{\pi}))\}<1-\epsilon<1$$

for some $\epsilon > 0$. Moreover, we also assume that samplings are independent to each other. So, $\forall l < \infty$, we have:

$$\mathbb{P}(\bigcap_{k=l}^{\infty}\{\mathcal{A}^{\pi}(S^{n_k}) \neq \mathcal{A}^{\hat{\pi}}(S^{n_k})\}) = \prod_{k=l}^{\infty}\mathbb{P}(\mathcal{A}^{\pi}(S^{n_k}) \neq \mathcal{A}^{\hat{\pi}}(S^{n_k})) \leq \lim_{n\to\infty} c(1-\epsilon)^n = 0$$

where c is some positive constant less than infinity. Therefore:

$$\mathbb{P}(\bigcap_{k=l}^{\infty}\{\mathcal{A}^{\pi}(S^{n_k}) \neq \mathcal{A}^{\hat{\pi}}(S^{n_k})\}) = 0.$$

This implies that we cannot find an $l < \infty$ such that $\hat{\pi}$ only takes l samples from A with positive probability. So $\hat{\pi}$ must be a convergent policy.

□

To discuss how MC perturb the suboptimal bound, we first need to define the expected one-step difference between π and $\hat{\pi}$. The one-step cost of $\hat{\pi}$ given $s \in \mathbb{S}$ is defined as:

$$C(s) = \sum_{x_k \in \chi} \mathbb{P}(\mathcal{A}^{\hat{\pi}}(s) = x_k)\left[\mathbb{E}[\min_i \mathbb{E}[\max_j \theta_{ij}|S^{n+1}]\big|S^n = s, x^n = x_k] - \min_{x' \in \chi}\mathbb{E}[\min_i \mathbb{E}[\max_j \theta_{ij}|S^{n+1}]\big|S^n\right]$$

where

$$\chi = \{1,\ldots,M\} \times \{1,\ldots,K\}.$$

Obviously, $V^{N-1,\pi}(s) = V^{N-1,\hat{\pi}}(s) - C(s)$. Let L be the number of samplings drawn in each MC estimate of $\mathcal{A}^{\pi}(.)$. C(s) is finite for any $s \in \mathbb{S}$ under our framework. Now we study the rate that C(s) shrinks to 0. For $x \in \chi$, according to central limit theorem we have

$$\frac{1}{L}\sum_{k=1}^{L} Y^{(k)}(x) \to \mathbb{E}\big[\min_i \mathbb{E}[\max_j \theta_{ij}|S^{n+1}]\big|S^n = s, x^n = x\big] \qquad \text{in the rate of O}(\frac{1}{\sqrt{L}})$$

where

$$Y^{(k)}(x;s) \text{ is the } k^{th} \text{ MC sample drawn to estimate } \mathbb{E}\big[\min_i \mathbb{E}[\max_j \theta_{ij}|S^{n+1}]\big|S^n = s, x^n = x\big]$$

From dominated convergence theorem, we can swap the limit sign and the integral:

$$\lim_{L\to\infty}\mathbb{P}\{\frac{1}{L}\sum_{k=1}^{L}Y^{(k)}(x) < \frac{1}{L}\sum_{k=1}^{L}Y^{(k)}(x')\} = \mathbb{E}\big[\lim_{L\to\infty}\mathbb{I}_{\{\frac{1}{L}\sum_{k=1}^{L}Y^{(k)}(x) < \frac{1}{L}\sum_{k=1}^{L}Y^{(k)}(x')\}}\big] \to 0$$
$$\text{in the rate of O}(\frac{1}{\sqrt{L}})$$

Therefore, for any $x \neq \mathcal{A}^{\pi}(s)$, we have:

$$\mathbb{P}\{\mathcal{A}^{\hat{\pi}}(s) = x\} = \prod_{x' \neq x}\mathbb{P}\{\frac{1}{L}\sum_{k=1}^{L}Y^{(k)}(x) < \frac{1}{L}\sum_{k=1}^{L}Y^{(k)}(x')\} \to 0 \qquad \text{in the rate of O}(\frac{1}{\sqrt{L}}).$$

From the same reasoning, we know that $\mathbb{P}\{\mathcal{A}^{\hat{\pi}}(s) = \mathcal{A}^{\pi}(s)\} \to 1$ in the rate of $O(\frac{1}{\sqrt{L}})$. So $\forall s \in \mathbb{S}$, $C(s) = O(\frac{1}{\sqrt{L}})$.

Now we have finished our preparation and begin the proof of Theorem 7.

Proof of Theorem 7:

From Proposition 6, we have:

$$\begin{aligned} V^{n,\hat{\pi}}(S^n) - C(S^n) - V^n(S^n) &\leq V^{N-1,\hat{\pi}}(S^n) - C(S^n) - V^n(S^n) \\ &= V^{N-1,\pi}(S^n) - V^n(S^n) \\ &= V^{N-1}(S^n) - V^n(S^n) \\ &\leq \max_{x^n,\ldots,x^{N-2}} \{ \sum_{k=n+1}^{N-1} \frac{1}{\sqrt{2\pi}} ||\tilde{\sigma}(\Sigma^k, \cdot)|| + \sqrt{2||\Sigma^k_{\cdot:,\cdot:}|| \log K} \}. \end{aligned}$$

The first line is because of benefits of measurement (Proposition 4) and the third equality is due to the definition of RKG. Now since $C(S^n) = O(\frac{1}{\sqrt{L}})$ and it is independent of N. According to the proof in theorem 8, we have $\mathbb{E}C(S^n) \to 0$ in the rate of $O(\frac{1}{\sqrt{L}})$. We can finish the proof.

□

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